

Module 7: Networking

At the end of this module, you will be able to:

- **Explain** the foundational role of an Amazon virtual private cloud (VPC) in AWS Cloud networking.
- **Identify** the networking components inside of a VPC and their purpose.
- **Differentiate** the options for VPC connectivity.
- **Describe** the layered network defense model inside a VPC.
- **List** the steps to troubleshoot common VPC network issues.
- **Configure** a VPC.

Topics:

- Networking and Amazon Virtual Private Cloud (Amazon VPC)
- Securing your network
- Troubleshooting networks on AWS

Knowledge check

Lab:

Lab 4: Configure an Amazon Virtual Private Cloud

Exercises/Project:

- Hands-on Activity 7: Troubleshoot a VPC
- Project: Troubleshooting Knowledge Base

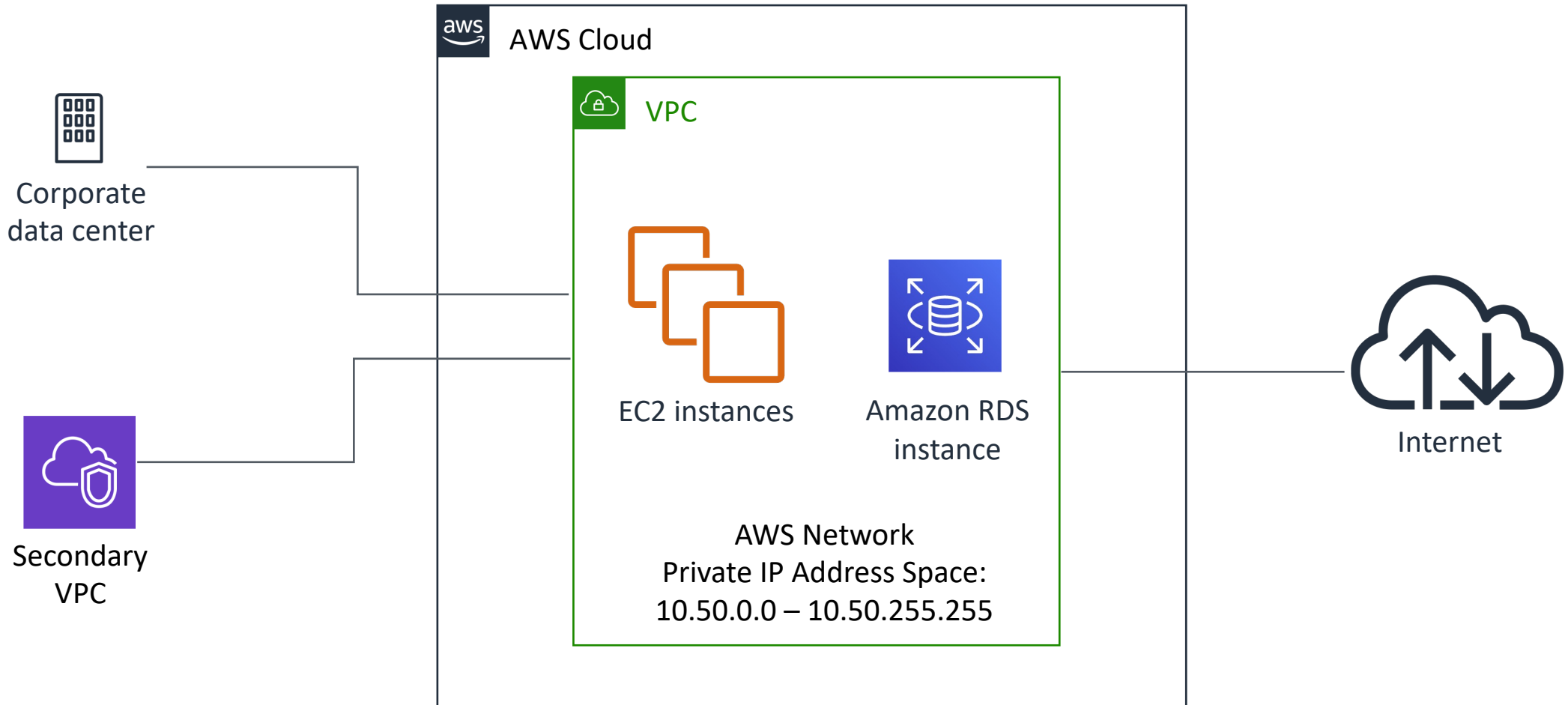
Troubleshooting knowledge base – Module 7 focus

- Have your copy of the troubleshooting knowledge base template open for editing
- **Create new entries** as you encounter troubleshooting situations
- **Categories of focus** for this module:
 - **Networking**
 - **Security and compliance**

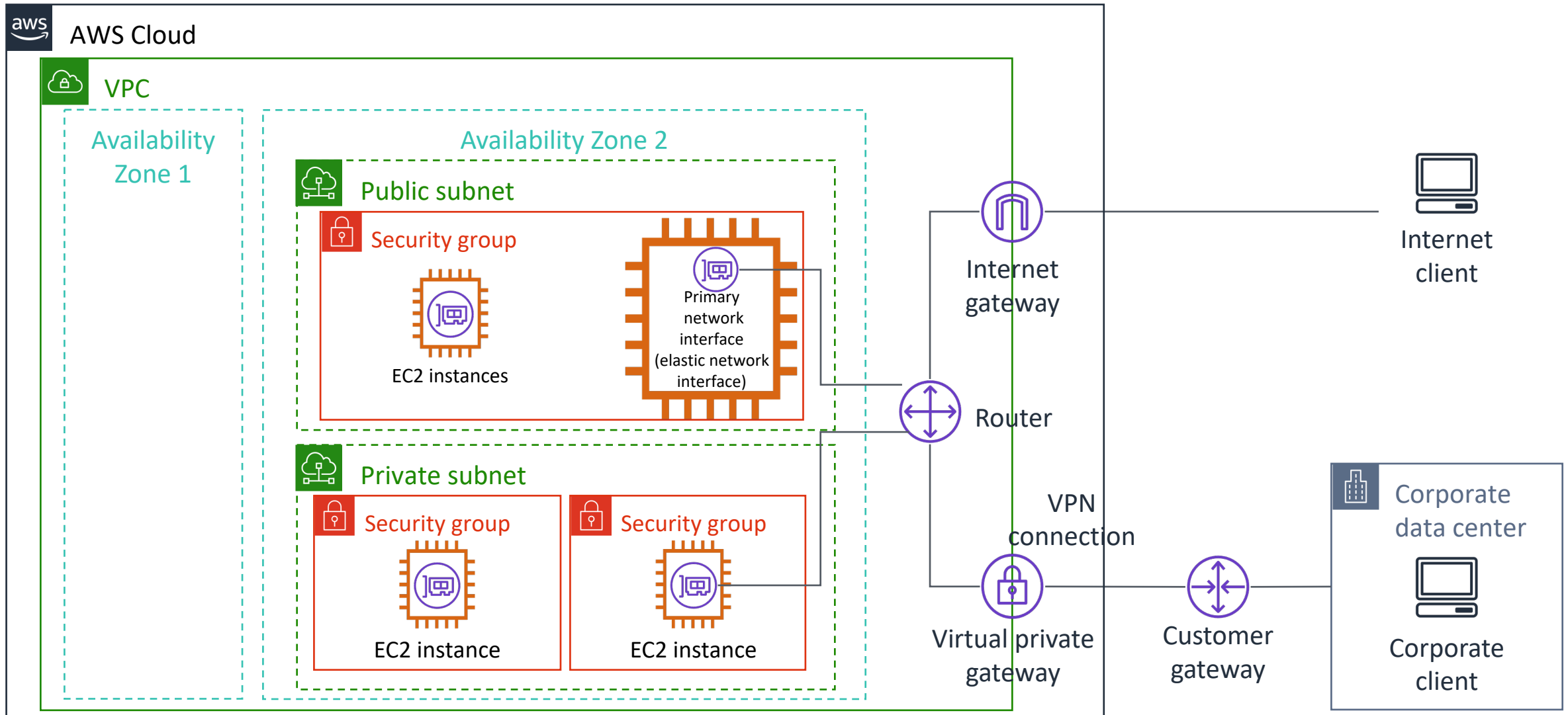
Module 7: Networking

AWS Cloud networking and Amazon Virtual Private Cloud (Amazon VPC)

AWS Cloud networking



AWS networking components





Amazon VPC

An **Amazon VPC** is a virtual network provisioned in a logically isolated section of the AWS Cloud:

- Supports logical separation with subnets
- Offers fine-grained security
- Supports optional hardware virtual private network (VPN)

IP addressing:

- Valid private IP address ranges defined by **RFC 1918**
- In an Amazon VPC, you can only define networks between /16 and /28.

Use non-overlapping IP address ranges.

Best Practice

Restrict VPCs to ranges defined in RFC1918 to avoid potential routing issues.

Example CLI command to create a VPC

Command:

```
$ aws ec2 create-vpc --cidr-block 10.0.0.0/16
```

Create VPC | VPC Management

https://console.aws.amazon.com/vpc/home

Services Resource Groups N. Virginia Support

VPCs > Create VPC

Create VPC

A VPC is an isolated portion of the AWS cloud populated by AWS objects, such as Amazon EC2 instances. You must specify an IPv4 address range for your VPC. Specify the IPv4 address range as a Classless Inter-Domain Routing (CIDR) block; for example, 10.0.0.0/16. You cannot specify an IPv4 CIDR block larger than /16. You can optionally associate an Amazon-provided IPv6 CIDR block with the VPC.

Name tag MomPopCafeVPC ⓘ

IPv4 CIDR block* 10.0.0.0/24 ⓘ

IPv6 CIDR block ☒ No IPv6 CIDR Block ⓘ
☐ Amazon provided IPv6 CIDR block

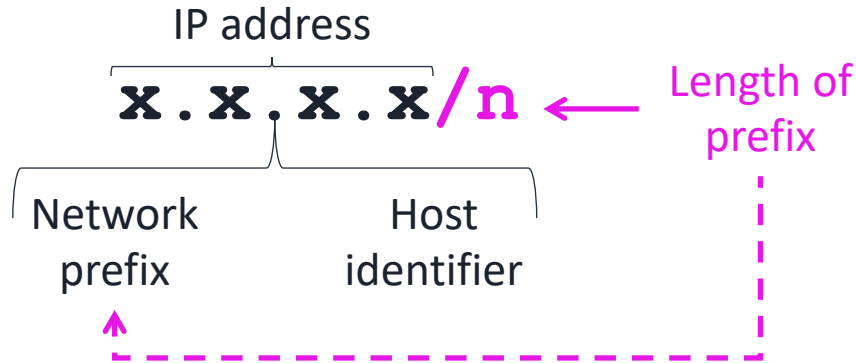
Tenancy Default ⓘ

* Required

Cancel Create

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Classless Inter-Domain Routing (CIDR) notation

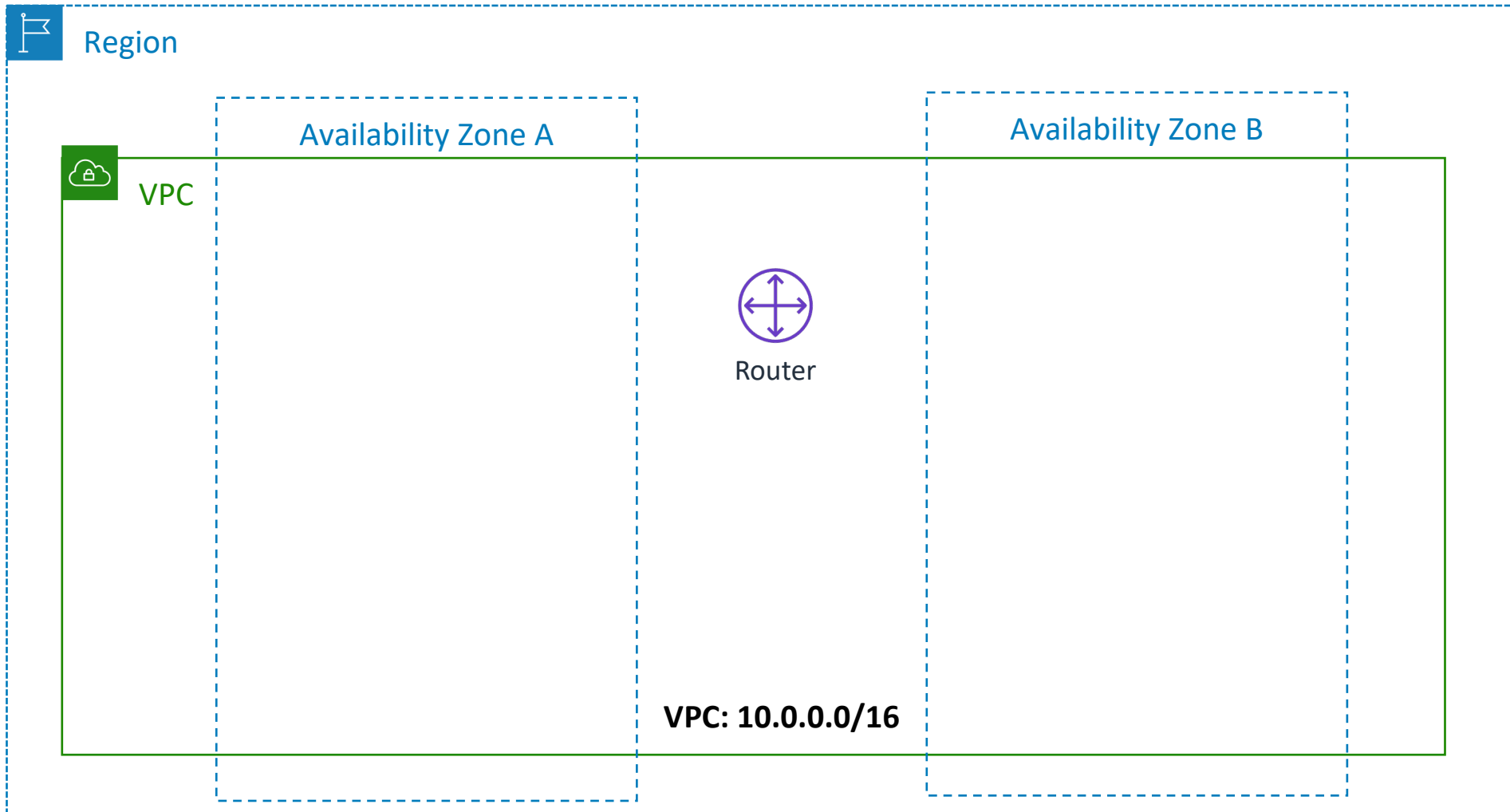


CIDR block:	Bit representation:	Corresponding address range:
10.50.1.0/24	00001010.00110010.00000001.xxxxxxxxxx	10.50.1.0 – 10.50.1.255
10.50.1.0/27	00001010.00110010.00000001.000xxxxxx	10.50.1.0 – 10.50.1.31
10.50.1.132/32	00001010.00110010.00000001.10000100	10.50.1.132 (single address)
0.0.0.0/0	xxxxxxxx.xxxxxxxxxx.xxxxxxxxxx.xxxxxxxxxx	0.0.0.0 – 255.255.255.255 (all addresses)

Amazon VPC reserved IP addresses

CIDR Block	Rationale
10.0.0.0	Network address.
10.0.0.1	Reserved by AWS for the VPC router address.
10.0.0.2	The IP address of the DNS server is always the base of the VPC network range plus two; however, VPC also reserves the base of each subnet range plus two.
10.0.0.3	Reserved by AWS for future use.
10.0.0.255	Network broadcast address. AWS does not support broadcast in a VPC, therefore this address is reserved.

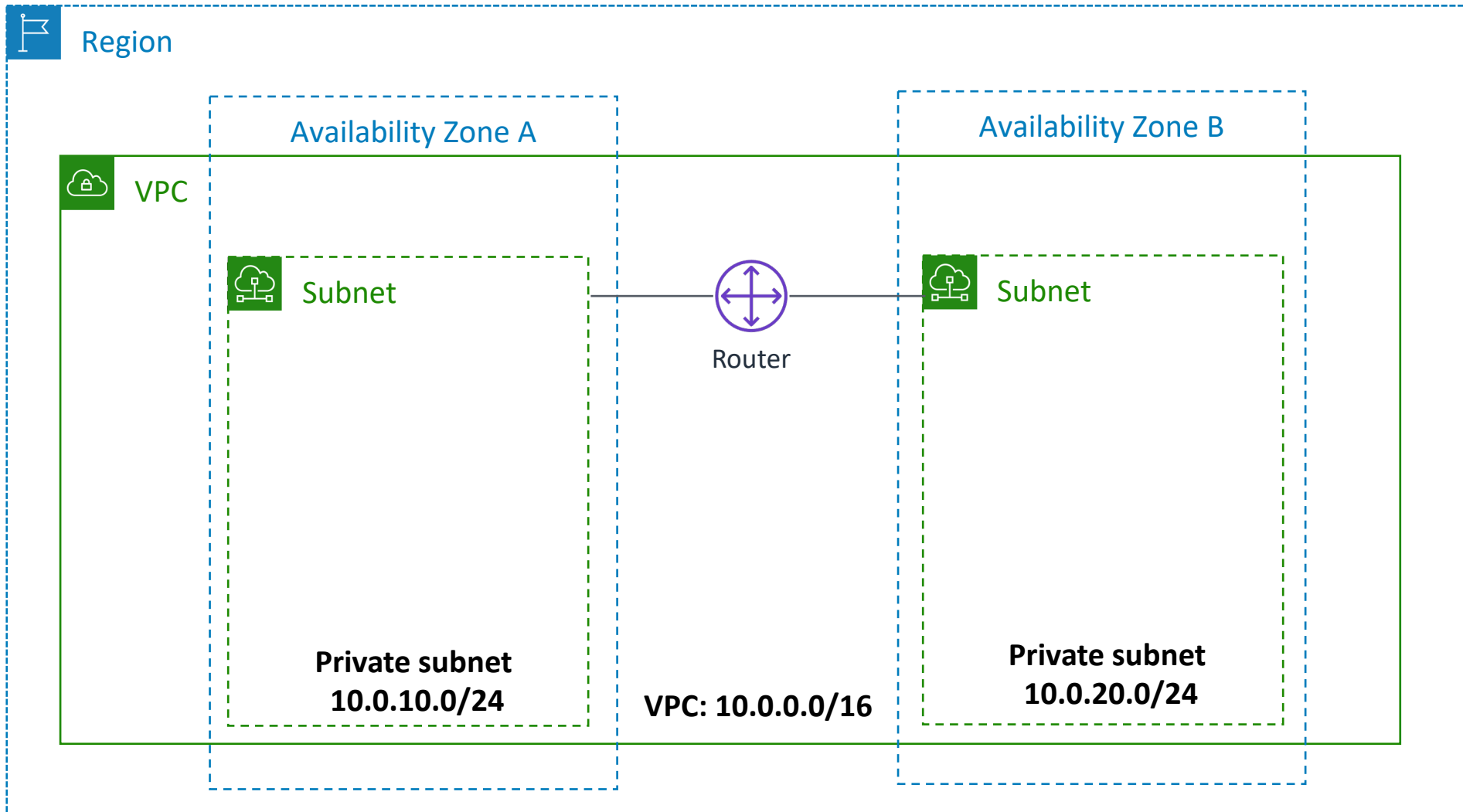
Amazon VPC characteristics



Default Route Table

Destination	Target
10.0.0.0/16	local

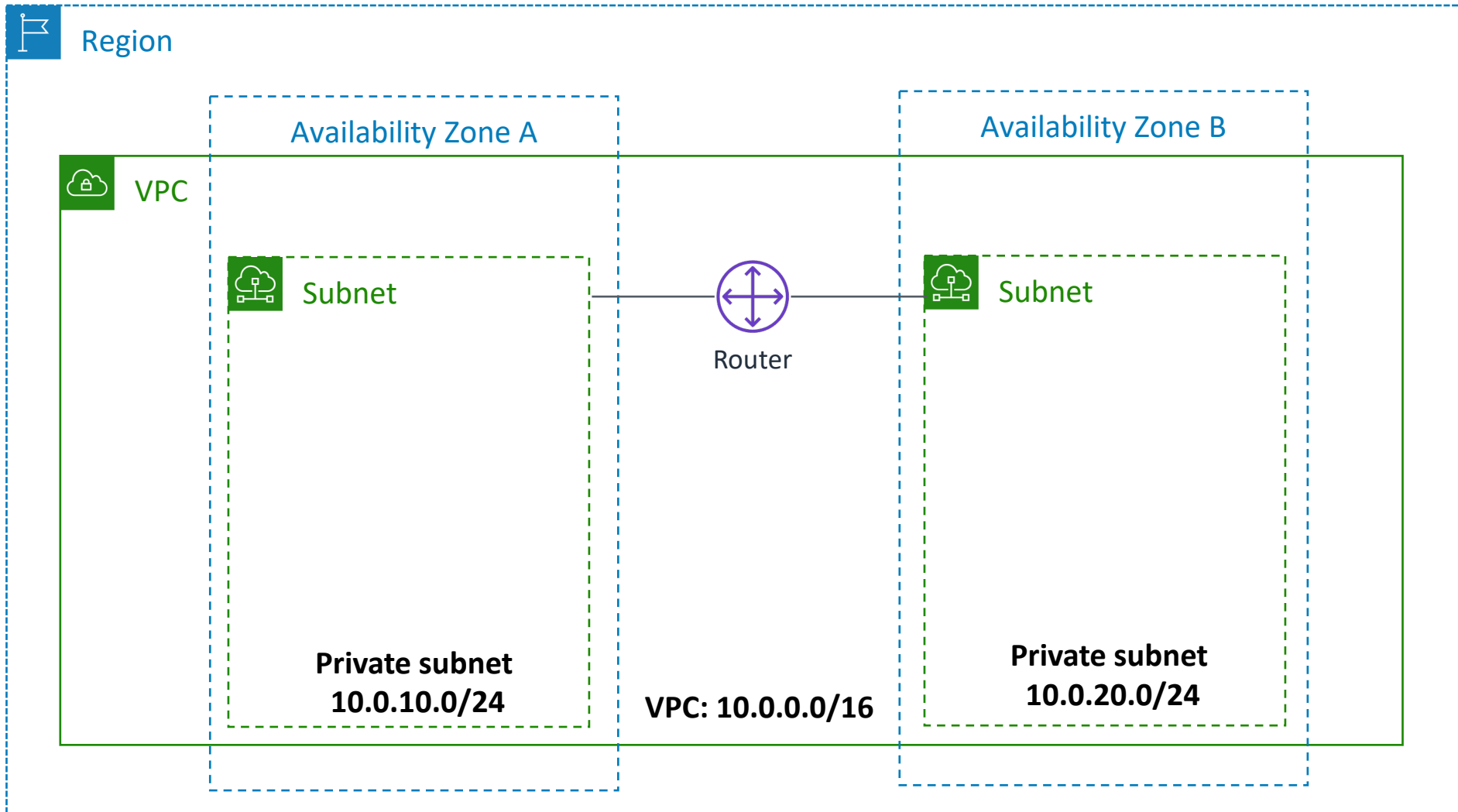
Amazon VPC components: Subnets



Default Route Table

Destination	Target
10.0.0.0/16	local

Amazon VPC components: Subnets continued



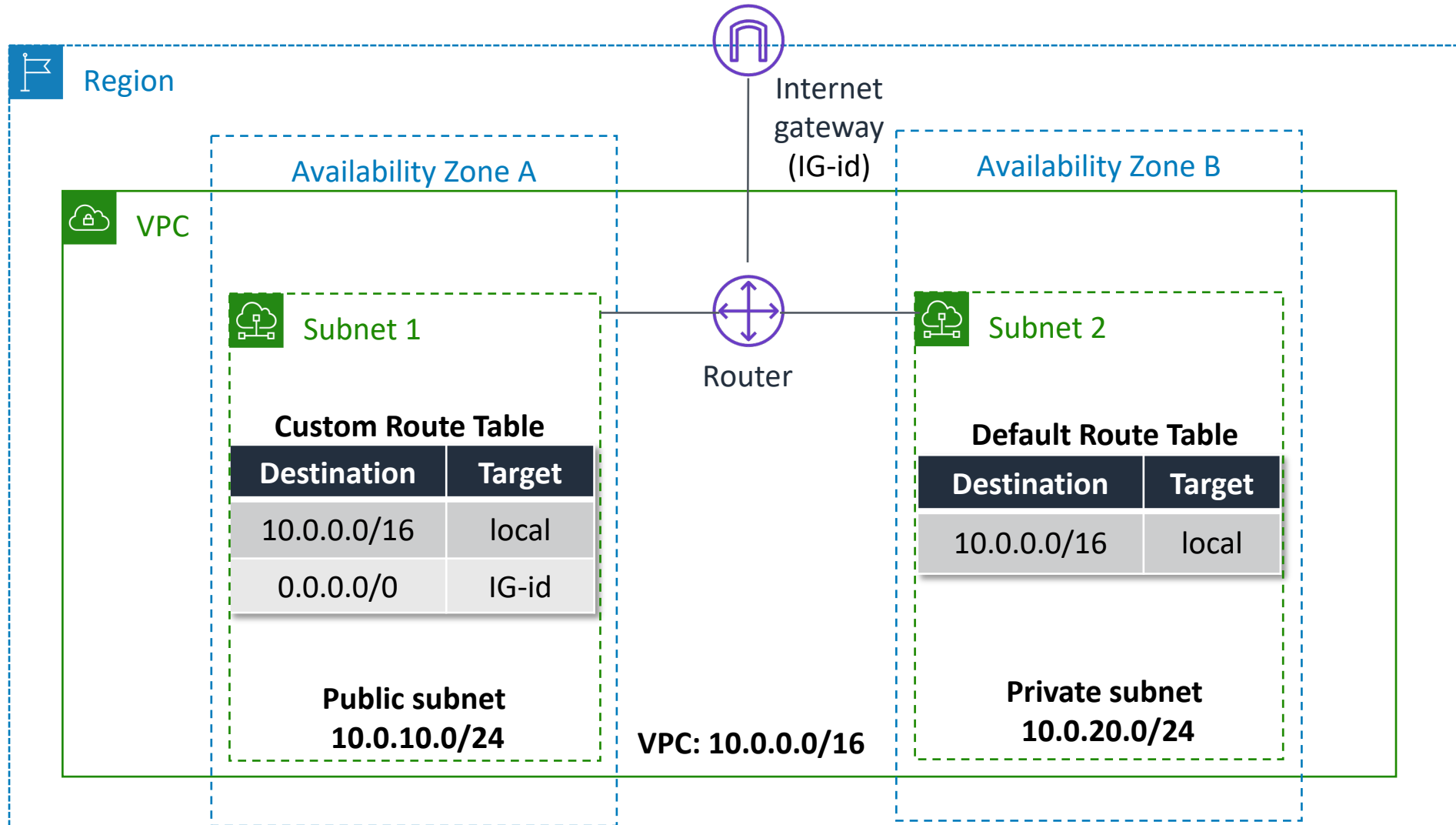
Default Route Table

Destination	Target
10.0.0.0/16	local

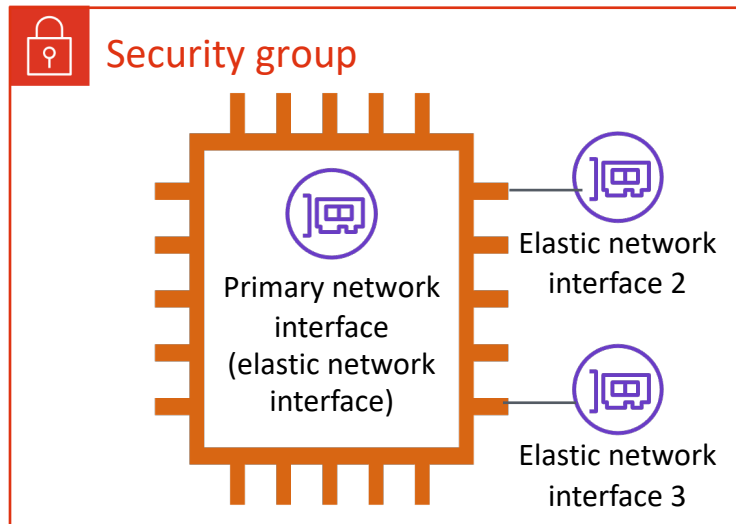
Auto-assign public IP subnet feature:

- Only instances launched in a default Amazon VPC receive a public IP address during creation.
- Instances created in custom VPCs require overriding the default setting to receive public IP addresses.
- Alternatively, configure the **Modify auto-assign IP settings** on subnets to automatically assign public IP addresses to instances.
- Unlike Elastic IP addresses, public IP addresses are not static.

Amazon VPC components: Route tables

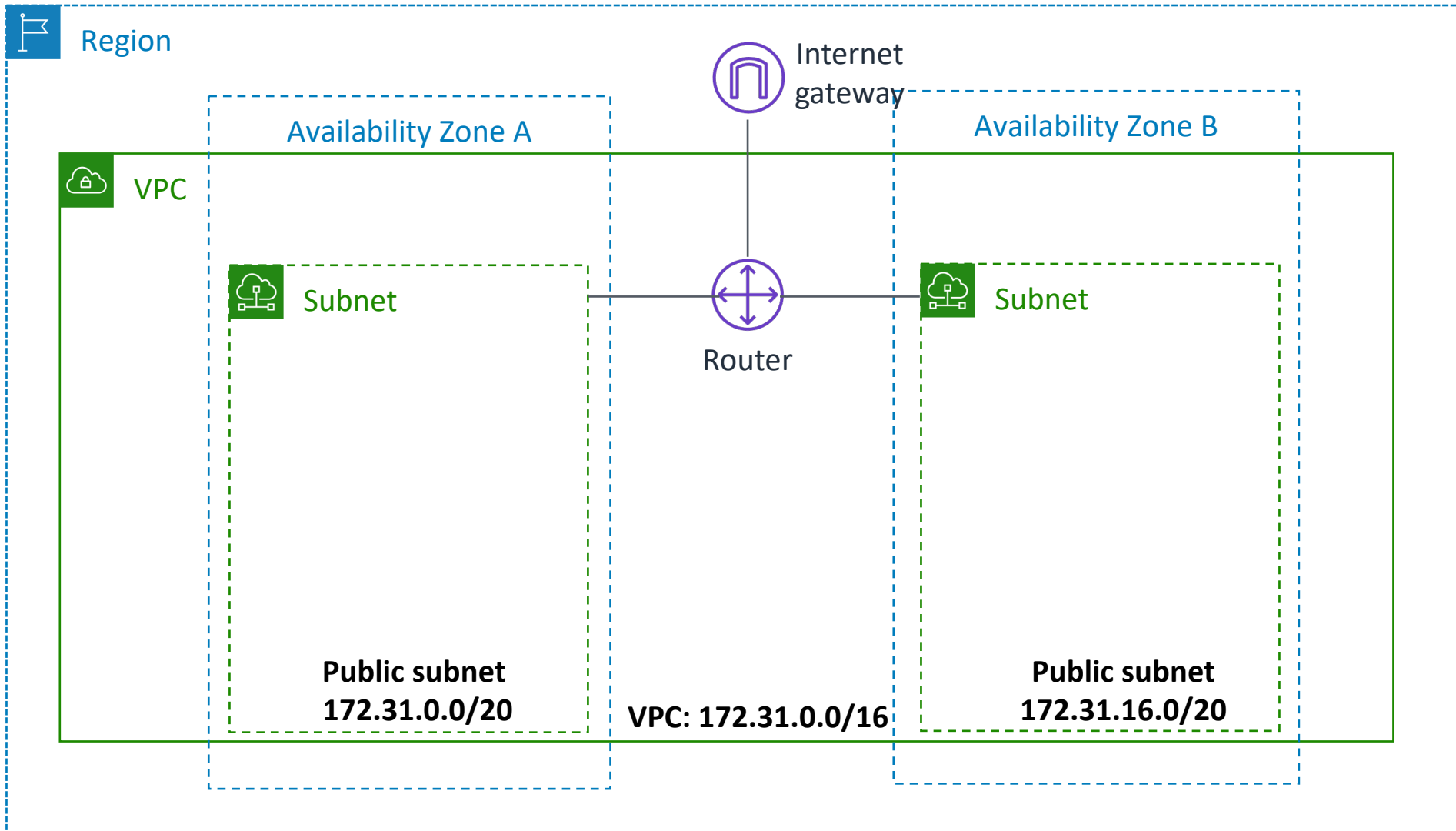


An **elastic network interface** is a virtual network interface (NIC) attached to an Amazon EC2 instance.



- Each instance has a **primary elastic network interface** that cannot be detached from it.
- You can attach additional network interfaces to an instance.
- Use cases include:
 - Using network and security appliances in a VPC
 - Connecting an instance to a service powered by AWS PrivateLink

Default VPC



Destination	Target
172.31.0.0/16	local
0.0.0.0/0	IG-id

DNS options for a VPC

Domain Name System (DNS) options for a VPC include:

DNS server provided by Amazon (Amazon Route 53 Resolver)

Your own DNS server

Amazon Route 53 private hosted zones

Key Takeaways



- Amazon VPC helps provision logically isolated networks on AWS.
- Subnets are classified as public or private based on their associated route tables.
- Security groups are stateful and NACLs are stateless firewalls. They are applied at elastic network interface and subnet levels respectively.

Module 7: Networking

Amazon VPC connectivity options

VPC connectivity scenarios and solutions

If You Need to:	Consider Using:	Solution Category:
Connect a private subnet to the internet	- NAT gateway - NAT instance	EC2 instance connectivity
Connect a VPC to another VPC	VPC peering	Amazon VPC to Amazon VPC
Connect a VPC to an external network	- AWS Site-to-Site VPN - AWS Direct Connect plus VPN - AWS VPN CloudHub	- Network to Amazon VPC - VPN connectivity
Connect a VPC to AWS services without leaving the AWS network	- AWS PrivateLink - VPC Gateway endpoint	- Amazon VPC to Amazon VPC - VPC Gateway endpoint
Connect a VPC to multiple VPCs and external networks	AWS Transit Gateway	- Network to Amazon VPC - Amazon VPC to Amazon VPC

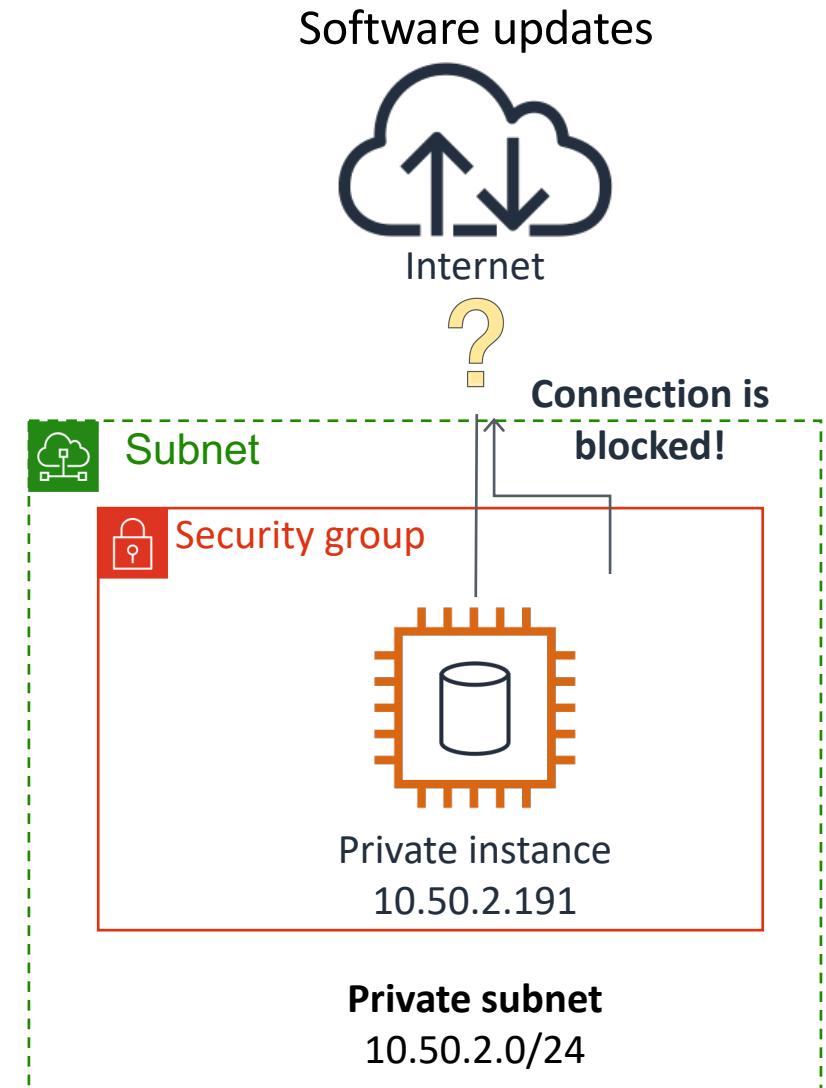
Network address translation (NAT)

Challenge:

Amazon EC2 instance in a private subnet needs to connect to the internet.

Solutions:

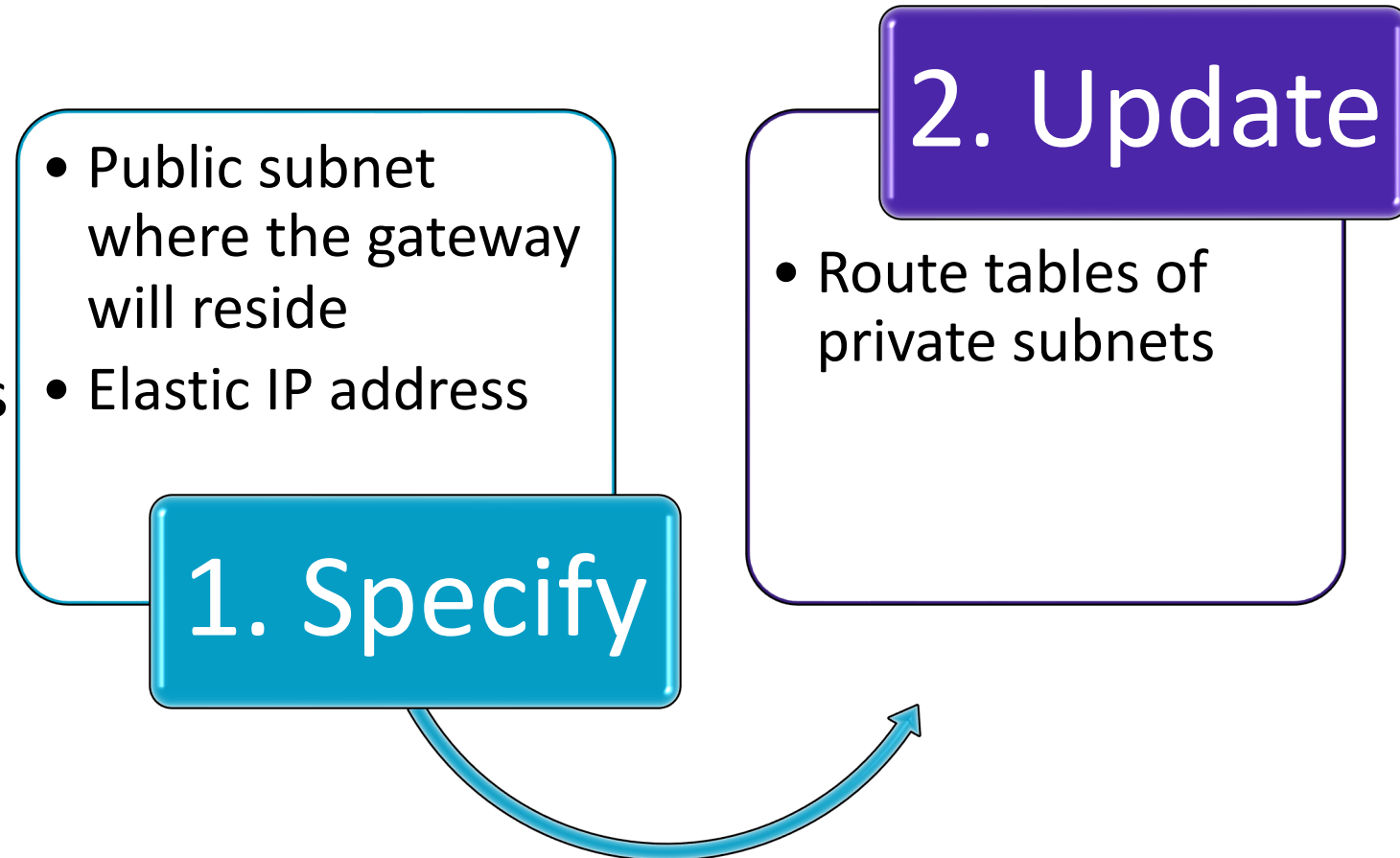
- ✓ NAT gateway
- ✓ NAT instance



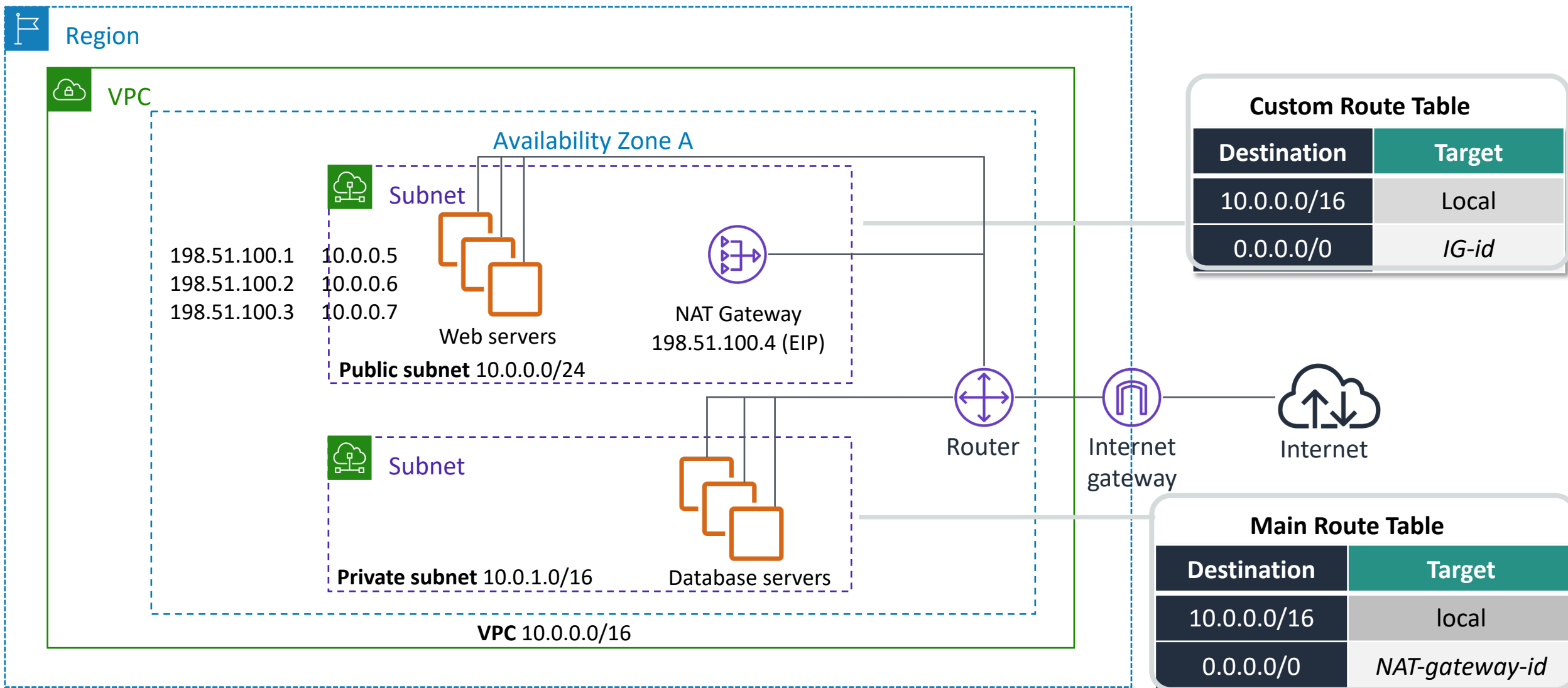
NAT gateway characteristics and creation steps

NAT gateway characteristics:

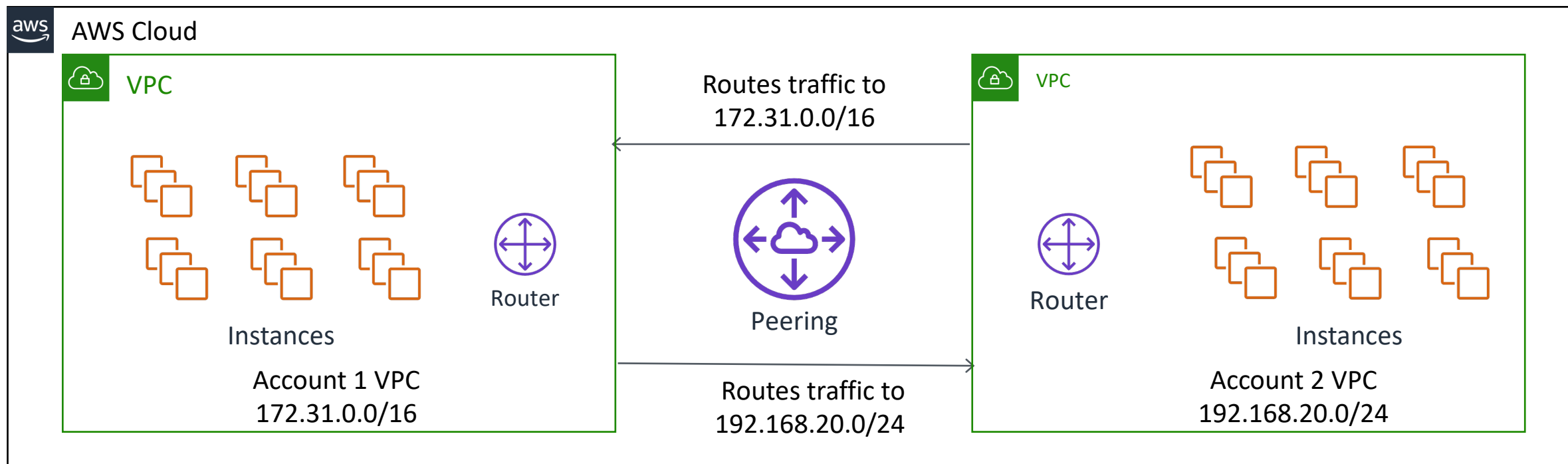
- AWS managed service
- Built-in redundancy within an availability zone.
- Requires an Elastic IP address
- Supports protocols:
 - TCP, UDP, and ICMP



Architecture of an Amazon VPC with a NAT gateway



Amazon VPC peering



Private Route Table

Destination	Target
171.31.0.0/16	Local
192.168.20.0/24	VPC peer

Private Route Table

Destination	Target
192.168.20.0/24	Local
172.31.0.0/16	PCX-XXXXXXX

Creating a VPC peering connection

To create a VPC peering connection:

1. Owner of requester VPC sends a VPC connection request.
2. Owner of acceptor VPC accepts the VPC connection request.
3. Both owners add route table entries in both participating VPCs.
4. If necessary, owners adjust security group rules in both participating VPCs.
5. If necessary, owners enable DNS hostname resolution for VPC connection.

CLI commands for VPC peering connection

Command:

```
aws ec2 create-vpc-peering-connection --vpc-id vpc-1a2b3c4d --peer-vpc-id vpc-11122233
```

Examples: AWS CloudFormation script entry

AWS CloudFormation script entry in JSON:

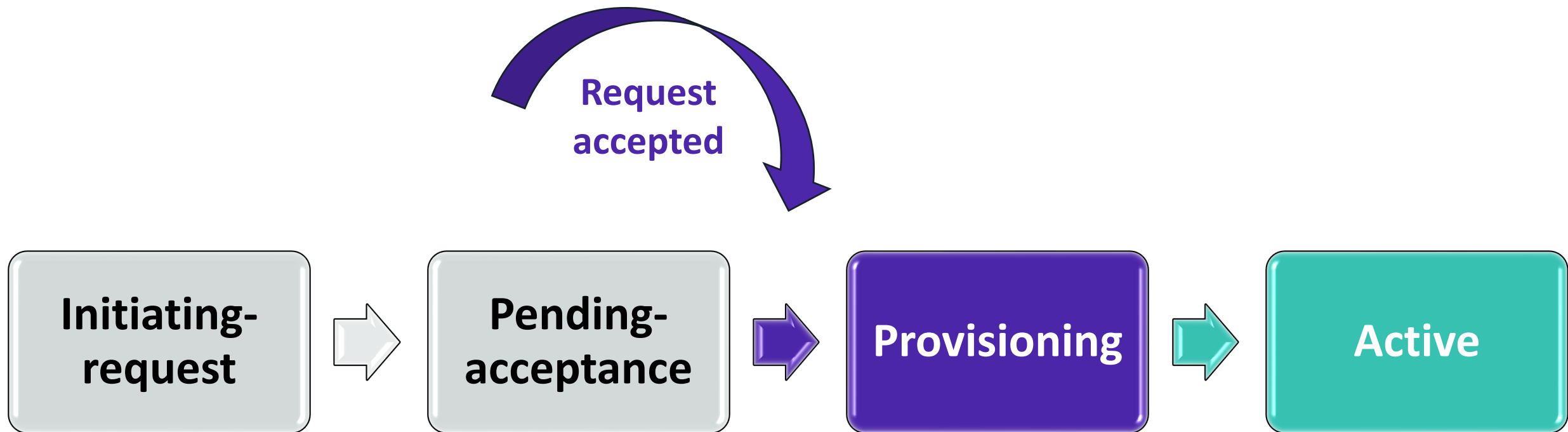
```
{
  "Type": "AWS::EC2::VPCPeeringConnection",
  "Properties": {
    "PeerVpcId": "vpc-1a2b3c4d",
    "VpcId": "vpc-11122233",
    "PeerOwnerId": "444455556666",
    "PeerRegion": "us-east-1",
    "PeerRoleArn": "arn:aws:ec2:us-east-1:444455556666"
  }
}
```

AWS CloudFormation script entry in YAML:

```
Type: AWS::EC2::VPCPeeringConnection
Properties:
  PeerVpcId: vpc-1a2b3c4d
  VpcId: vpc-11122233
  PeerOwnerId: 444455556666
  PeerRegion: us-east-1
  PeerRoleArn: arn:aws:ec2:us-east-1:444455556666
```

VPC peering connection creation states

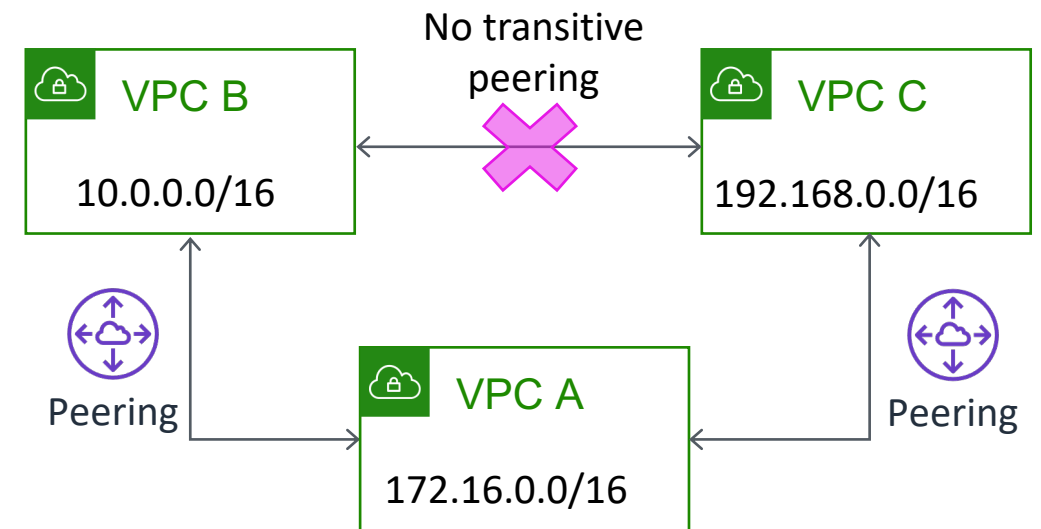
VPC peering connection creation states include:



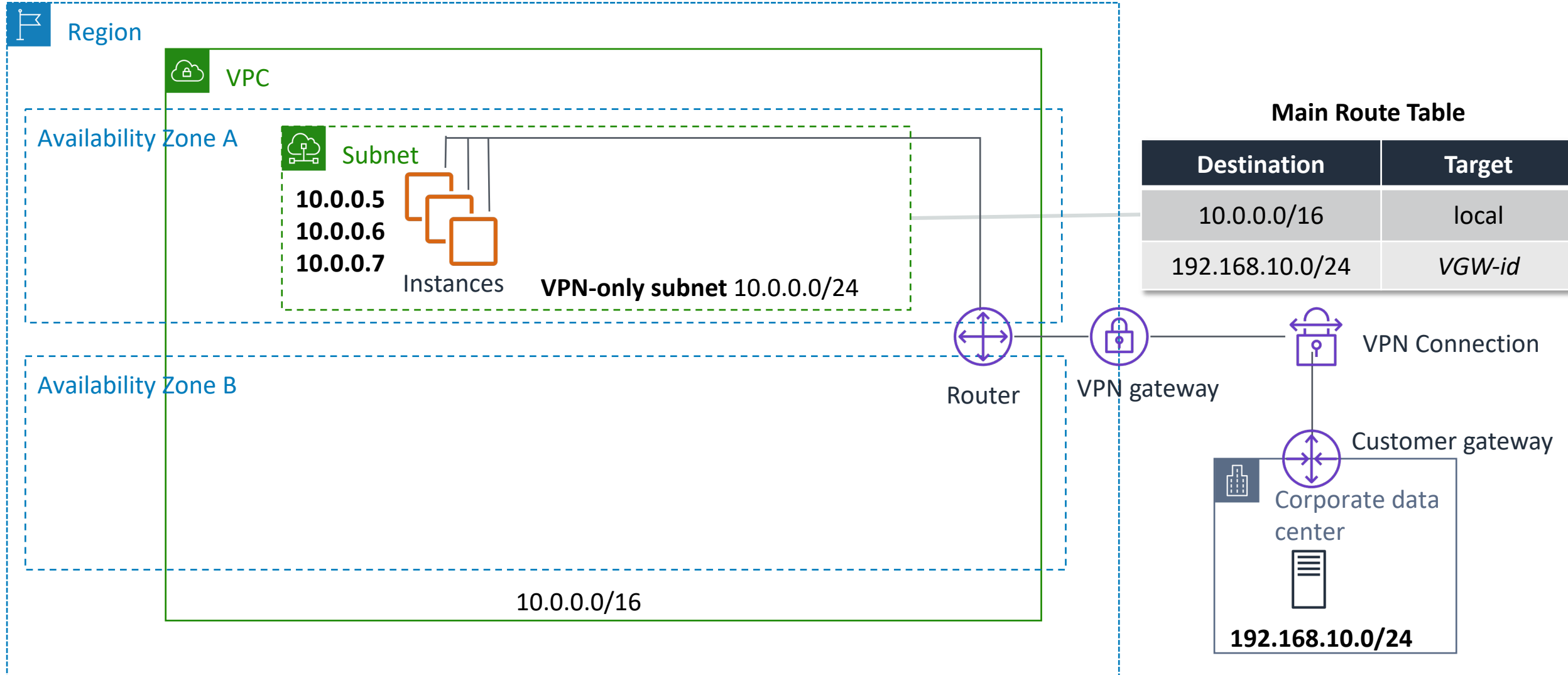
VPC peering limitations

The VPC peering limitations include:

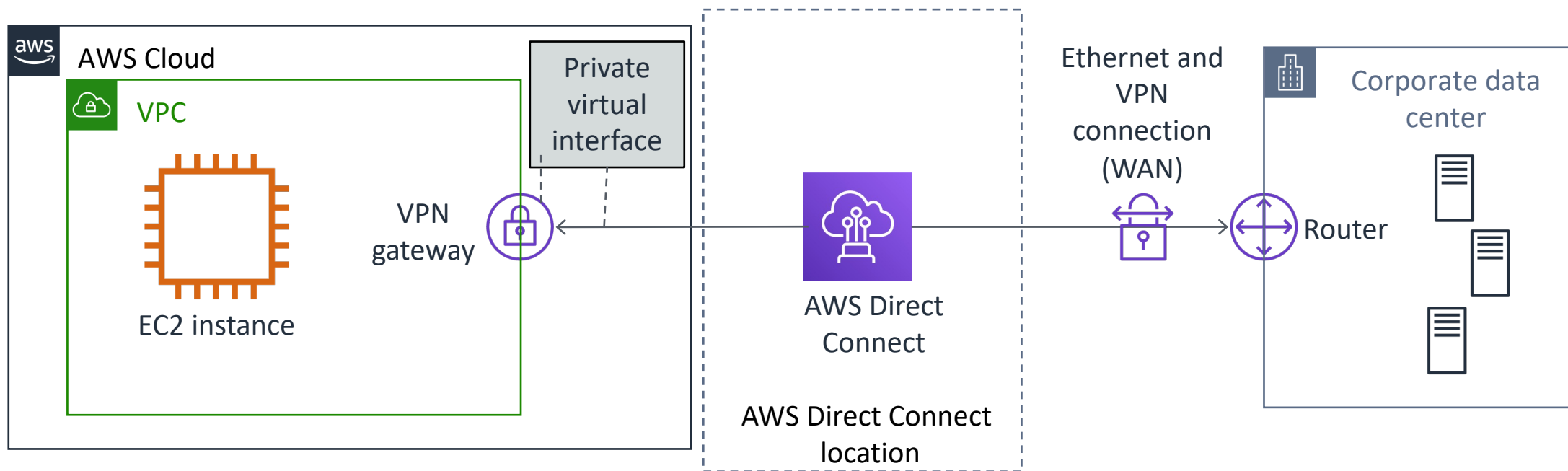
- No overlapping IP address ranges.
- No transitive peering, edge routing, or internet gateway access.
- No NAT routing between VPCs.
- No DNS lookup resolution of private IP addresses.
- Supports cross-referencing security groups across Regions.

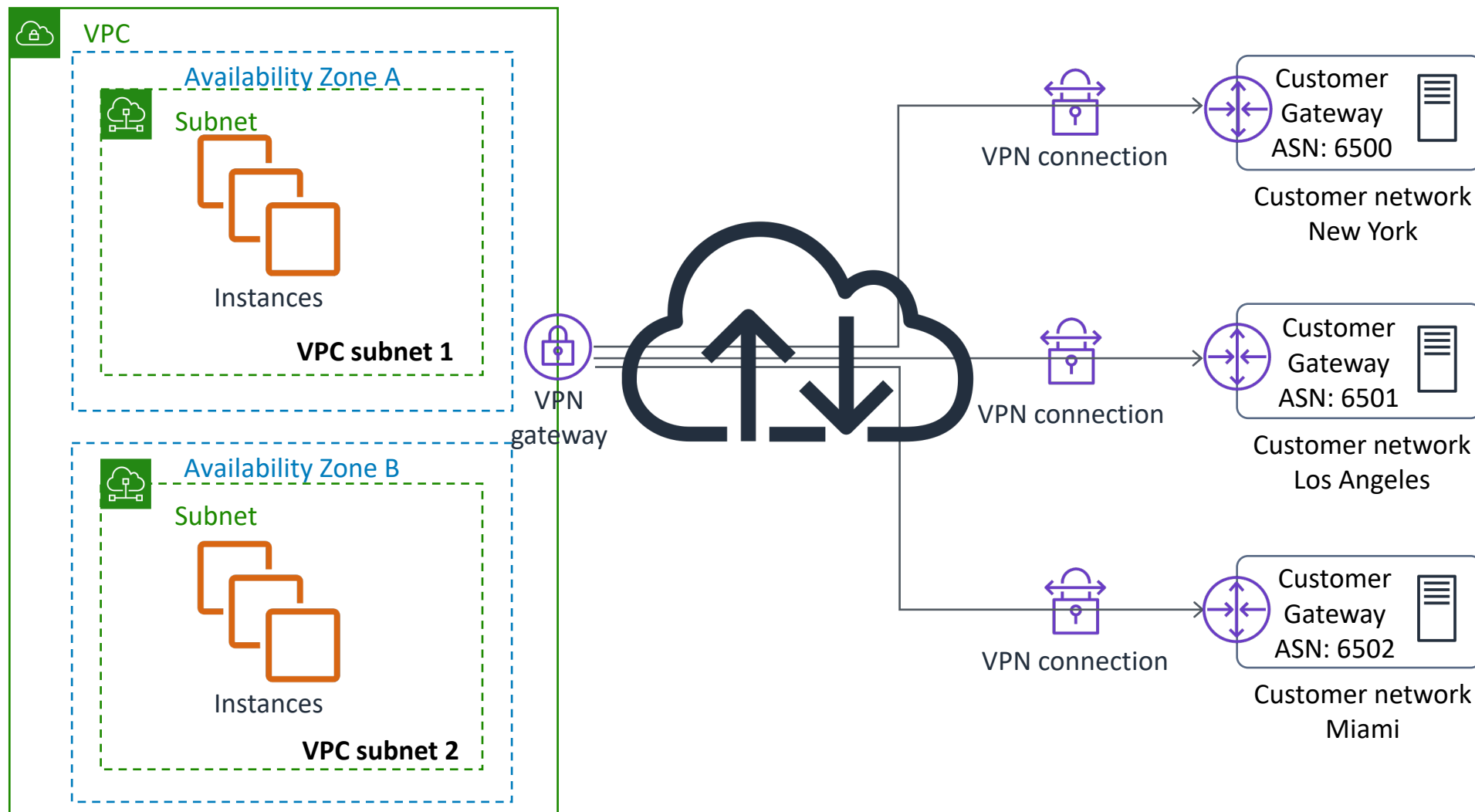


AWS site-to-site VPN

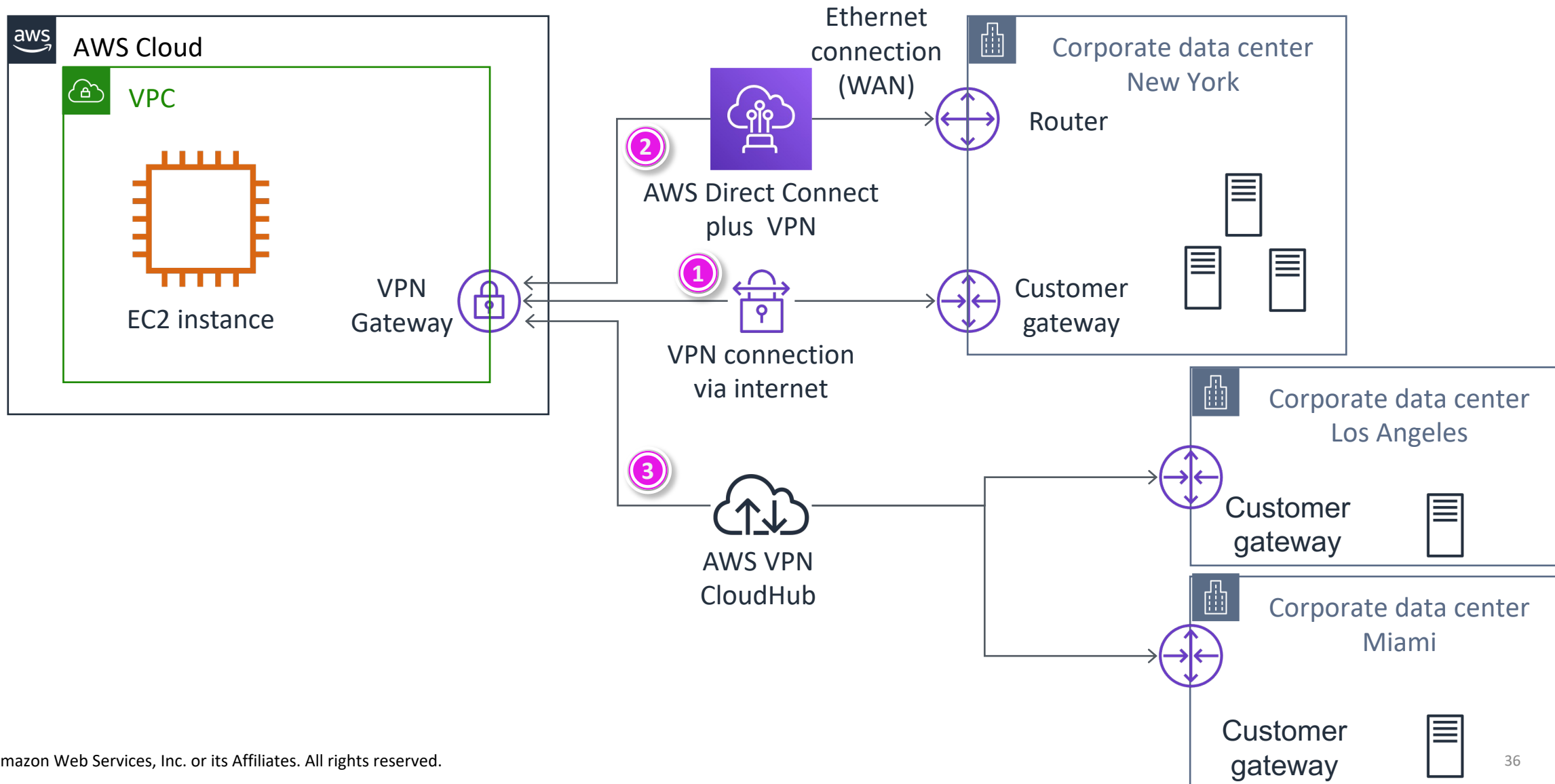


AWS Direct Connect plus VPN

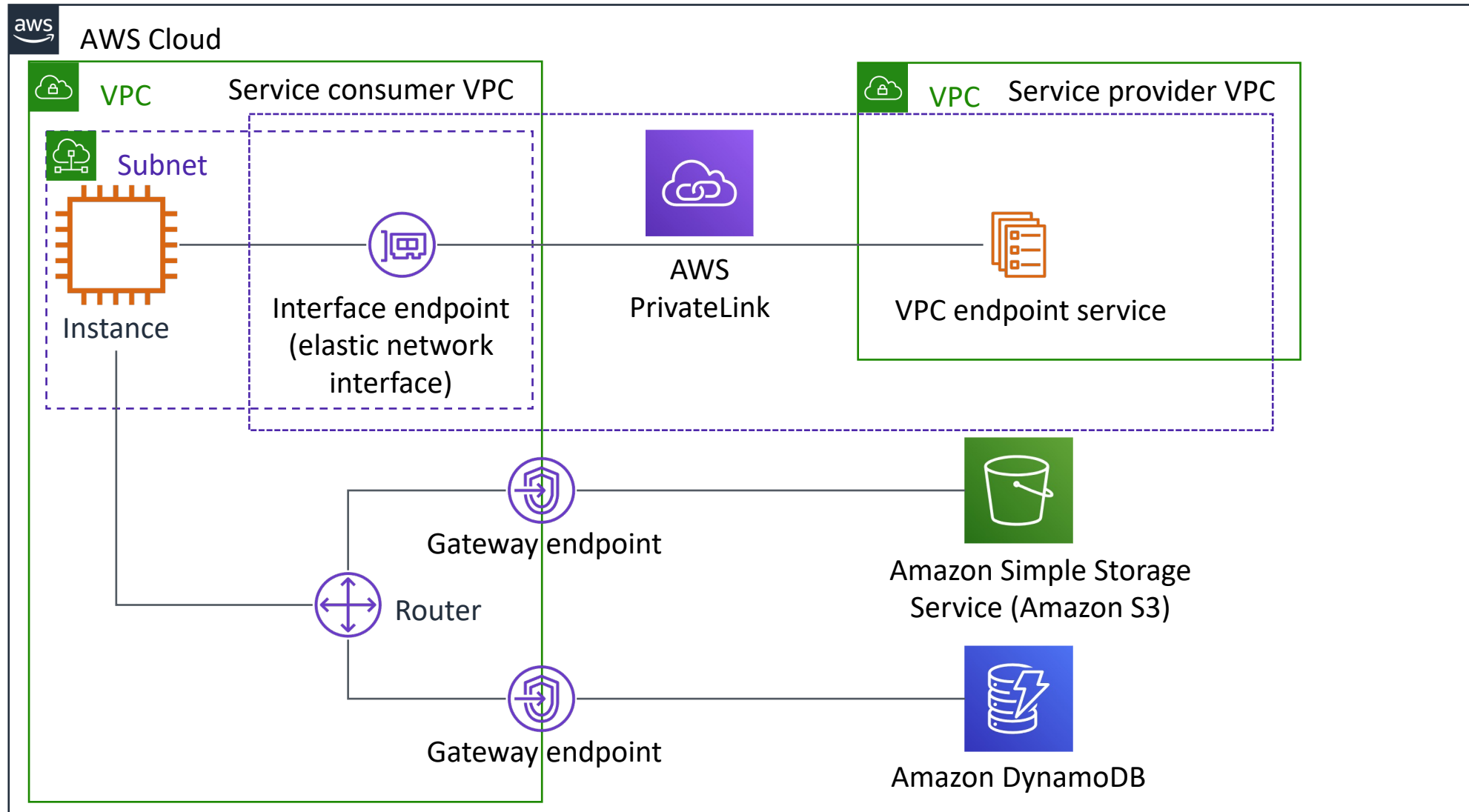




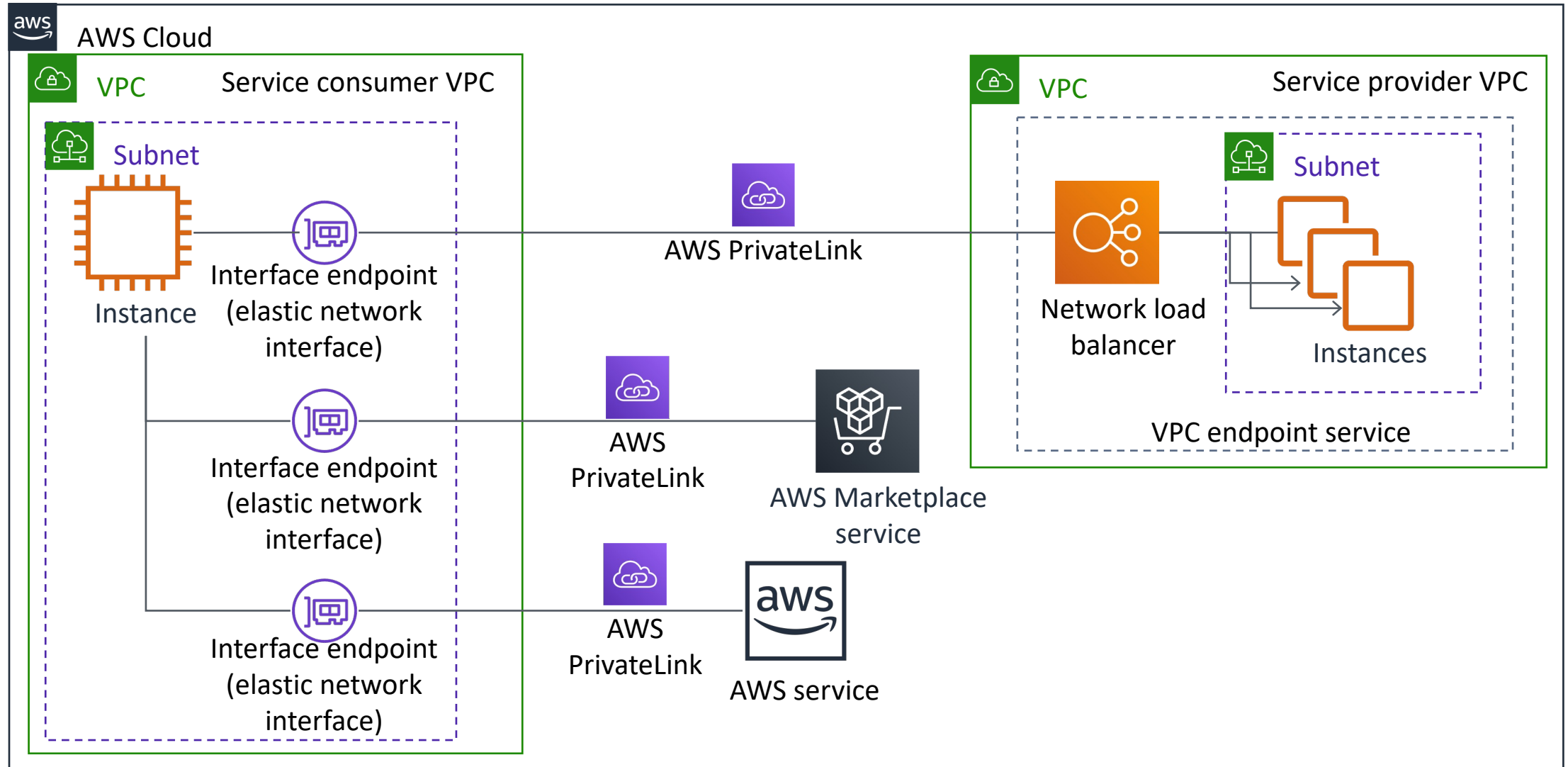
VPN connection options



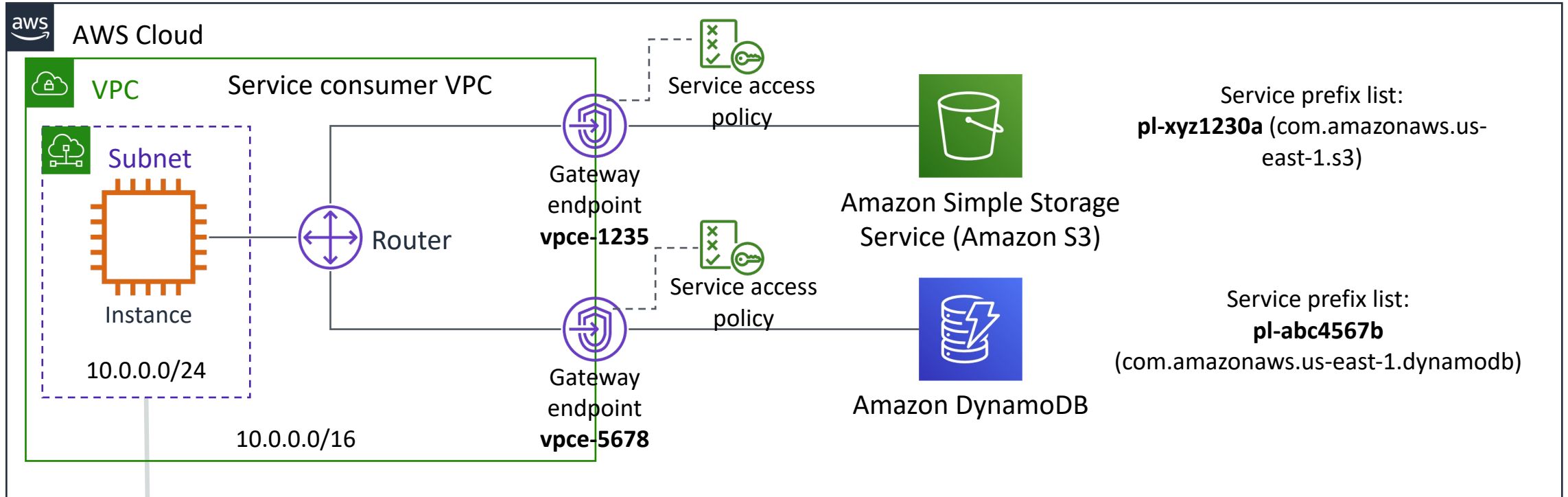
VPC endpoints



AWS PrivateLink interface endpoint



Gateway endpoint



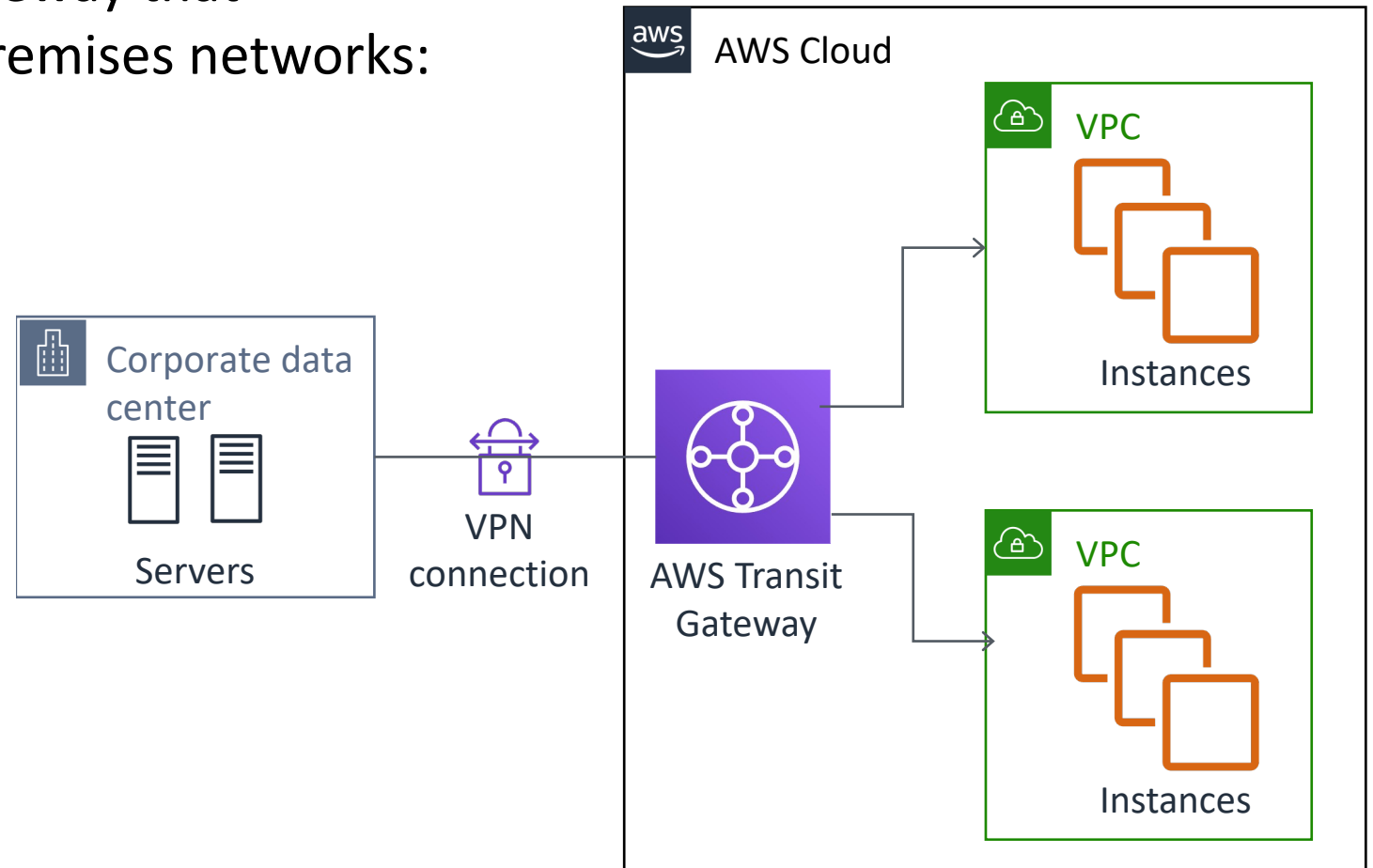
Subnet Route Table

Destination	Target
10.0.0.0/16	Local
pl-xyz1230a	vpce-1235
pl-abc4567b	vpce-5678

AWS Transit Gateway

AWS Transit Gateway is a single gateway that connects thousands VPCs and on-premises networks:

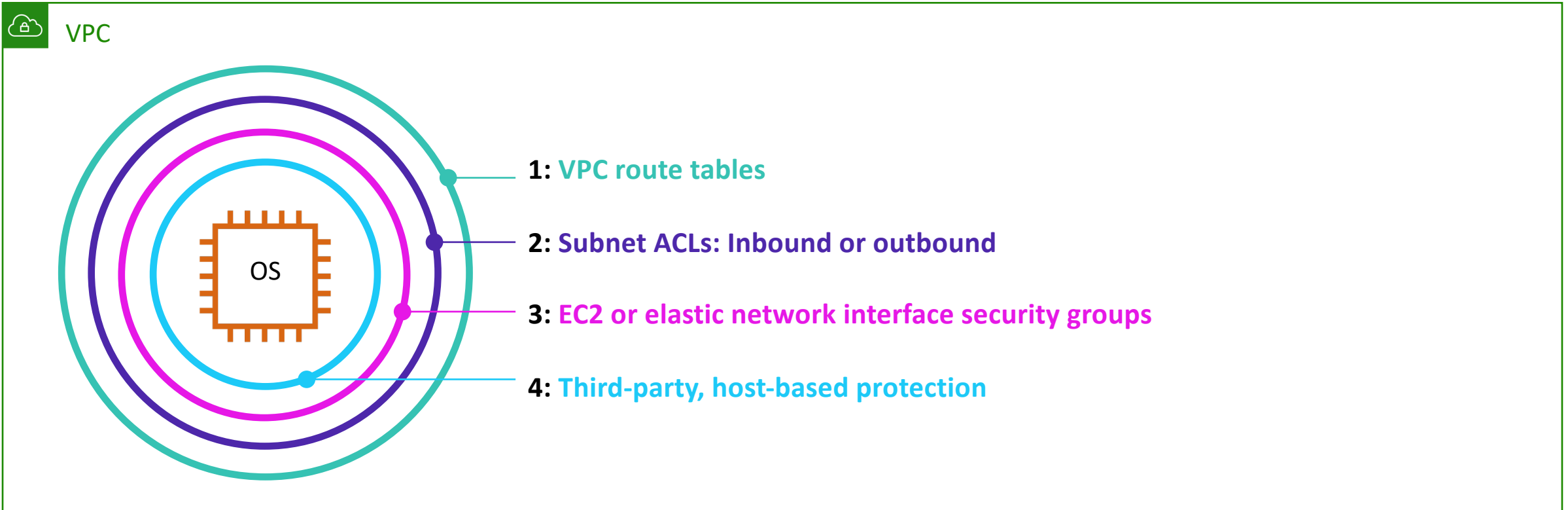
- Attaches to VPCs and VPNs
- Supports route propagation from the route tables of the attached VPCs and VPN connections



Module 7: Networking

Securing your network

Layered network defense for VPCs



Security groups:

- Used to allow traffic to or from an elastic network interface
- Configured by default to:
 - **Deny** all inbound traffic
 - **Allow** all outbound traffic
- **Stateful**: If rules allow traffic to flow in one direction, responses can automatically flow in the opposite direction.
- Usually administered by application developers

Outbound Rules

Type	Protocol	Port Range	Source
All traffic	All	All	0.0.0.0/0



Inbound Rules

Type	Protocol	Port Range	Source
HTTP	TCP	80	0.0.0.0/0
SSH	TCP	22	0.0.0.0/0
MYSQL/Aurora	TCP	3306	0.0.0.0/0

Network access control lists (network ACL):

- Allow or deny traffic in and out of subnets
- Default network ACL:
 - **Allows** all inbound and outbound traffic
- **Stateless**: Even if rules allow traffic to flow in one direction, you have to explicitly allow responses to flow in the opposite direction.
- Hardens security as a secondary level of defense at the subnet level



Network access
control list

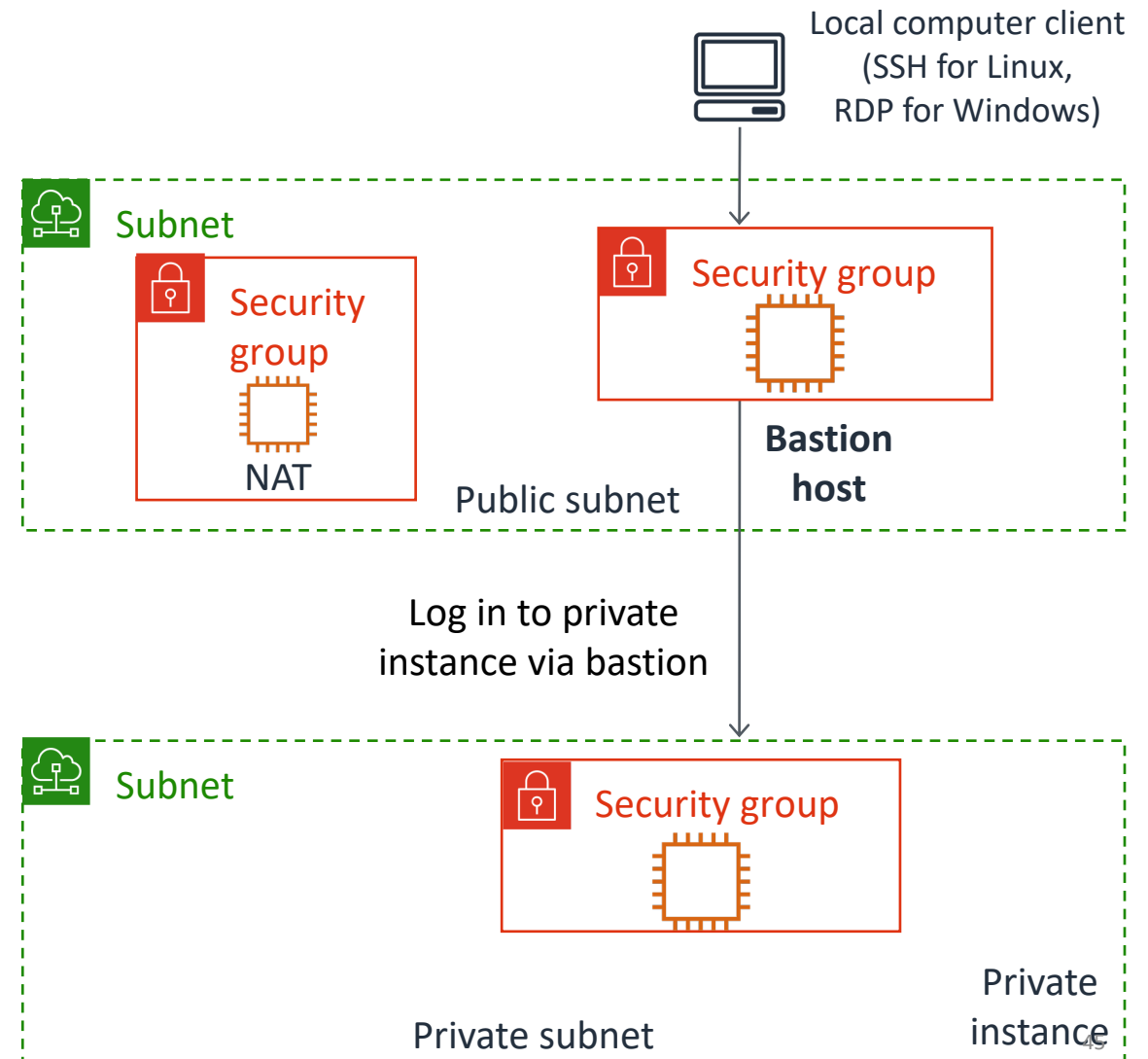
Inbound Rules for Default Network ACL

Rule #	Type	Protocol	Port Range	Source	Allow or Deny
100	ALL Traffic	ALL	80	0.0.0.0/0	ALLOW
*	SSH	TCP	22	0.0.0.0/0	DENY

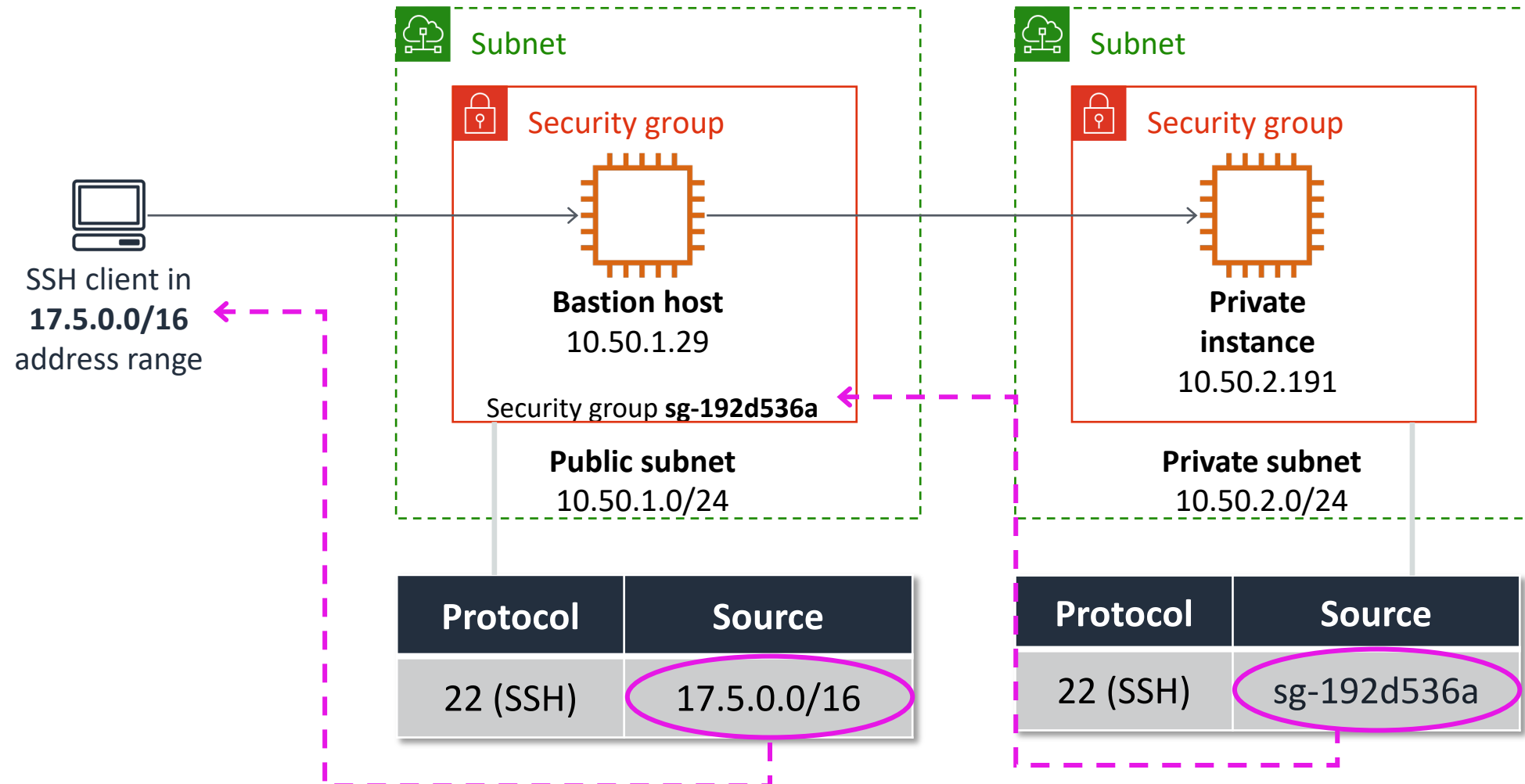
Bastion host

The bastion host:

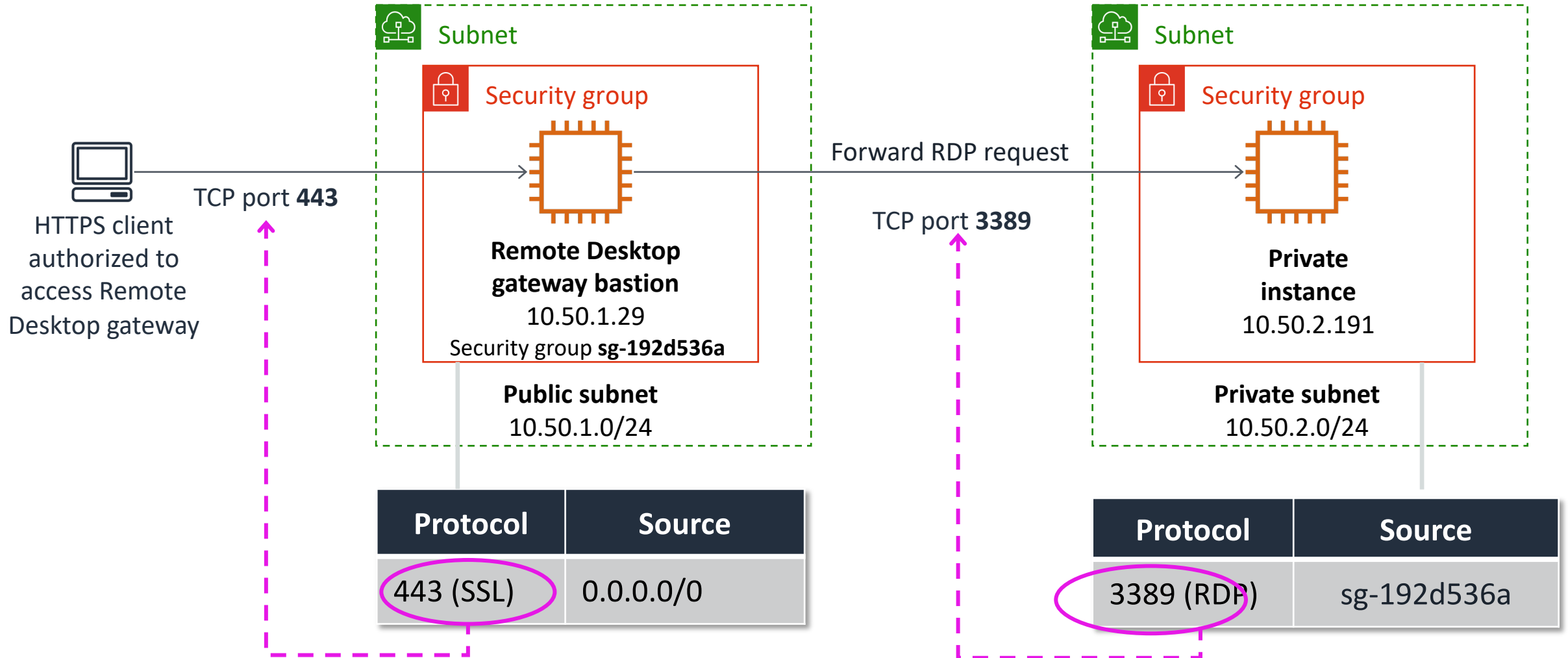
- Provides public-subnet-to-private-subnet access.
 - Jump point to gain access to a private subnet
- Do not store private keys.
 - For Linux instances, use the agent forwarding feature of the SSH client to specify the key.
 - For Windows instances, decrypt password using the key in the Amazon EC2 console and then add the instance to a domain.



Linux bastion host security groups



Remote desktop gateway as Windows bastion host



Module 7: Networking

Troubleshooting networks on AWS

Common troubleshooting tasks

1. Verify that the instance is up and running.
 - Check that it has passed both the **System Status** and **Instance Status** checks.
2. Verify that the security groups associated with the instance allow connections for the required protocols and ports.
3. Verify that the network ACLs associated with the subnet allow traffic from the necessary ports and protocols.
4. Verify that the route table associated with the subnet has destination rules that point to the appropriate targets.



Troubleshooting instance connections

Common troubleshooting tasks

- Verify that the instance is running.
- Verify security groups rules.
- Verify network ACL rules.
- Verify route table rules.

If you cannot connect to the instance via the internet:

- Verify the public IP address or DNS name you are using is correct.
- Verify the instance has a public IP address or Elastic IP address.
- Verify that an internet gateway is attached to the instance's VPC.
- Verify that the route table of the instance's subnet has a route rule for the destination 0.0.0.0/0 via the internet gateway.

Troubleshooting SSH connections

Common troubleshooting tasks

- Verify that the instance is running.
- Verify port 22 is open in the security group.
- Verify network ACL rules.
- Verify route table rules.

If you cannot connect to an instance using SSH:

- Verify the instance connection credentials:
 - User name
 - Instance private key

Common troubleshooting tasks

- Verify that the instance is running.
- Verify security groups rules.
- Verify network ACL rules.
- Verify route table rules.

If the NAT configuration does not work:

- NAT instance or NAT gateway:
 - Verify that the route table has a route to the NAT instance or NAT gateway
- NAT instance:
 - Check to see if the **Source/Dest Check** is disabled.
 - Restart the NAT instance.
 - Check the **inbound** security group rules.

Common troubleshooting tasks

- Verify that the instance is running.
- Verify security groups rules.
- Verify network ACL rules.
- Verify route table rules.

If you cannot reach resources in a peered network:

- Make sure the peering request was approved.
- **Check the security group rules:** Make sure to use the CIDR block range in VPC A to allow access from VPC B or use VPC A's security group ID.
- **Check the network ACLs:** Are you forbidding all external traffic?

In summary, this module discussed:

- The foundational role of a virtual private cloud (VPC) in AWS Cloud networking.
- The networking components inside of a VPC and their purpose.
- The options for VPC connectivity.
- The layered network defense model inside a VPC.
- The steps to troubleshoot common VPC network issues.
- How to configure a VPC.

To finish this module:

Complete:  **Knowledge check**

Test Strategies



Module 7 Scenario:

The automotive industry is now developing a specific unmanned ground vehicle (UGV) to be used by the military that provides alerts to the operator of hazardous conditions, loss of connection, or vehicle damage. They know they will need to align with the National Institute of Standards and Technology (NIST) guidelines 800-171 regarding the vehicle's security strength to protect its location at all times. The application used by the UGV will be hosted in a private subnet.

What needs to be updated after establishing a network address translation (NAT) gateway for the unmanned vehicle application so operators can connect and retrieve location data securely? Select three responses.

Module 7 Scenario:

The automotive industry is now developing a specific unmanned ground vehicle (UGV) to be used by the military that provides alerts to the operator of hazardous conditions, loss of connection, or vehicle damage. They know they will need to align with the National Institute of Standards and Technology (NIST) guidelines 800-171 regarding the vehicle's security strength to protect its location at all times. The application used by the UGV will be hosted in a private subnet.

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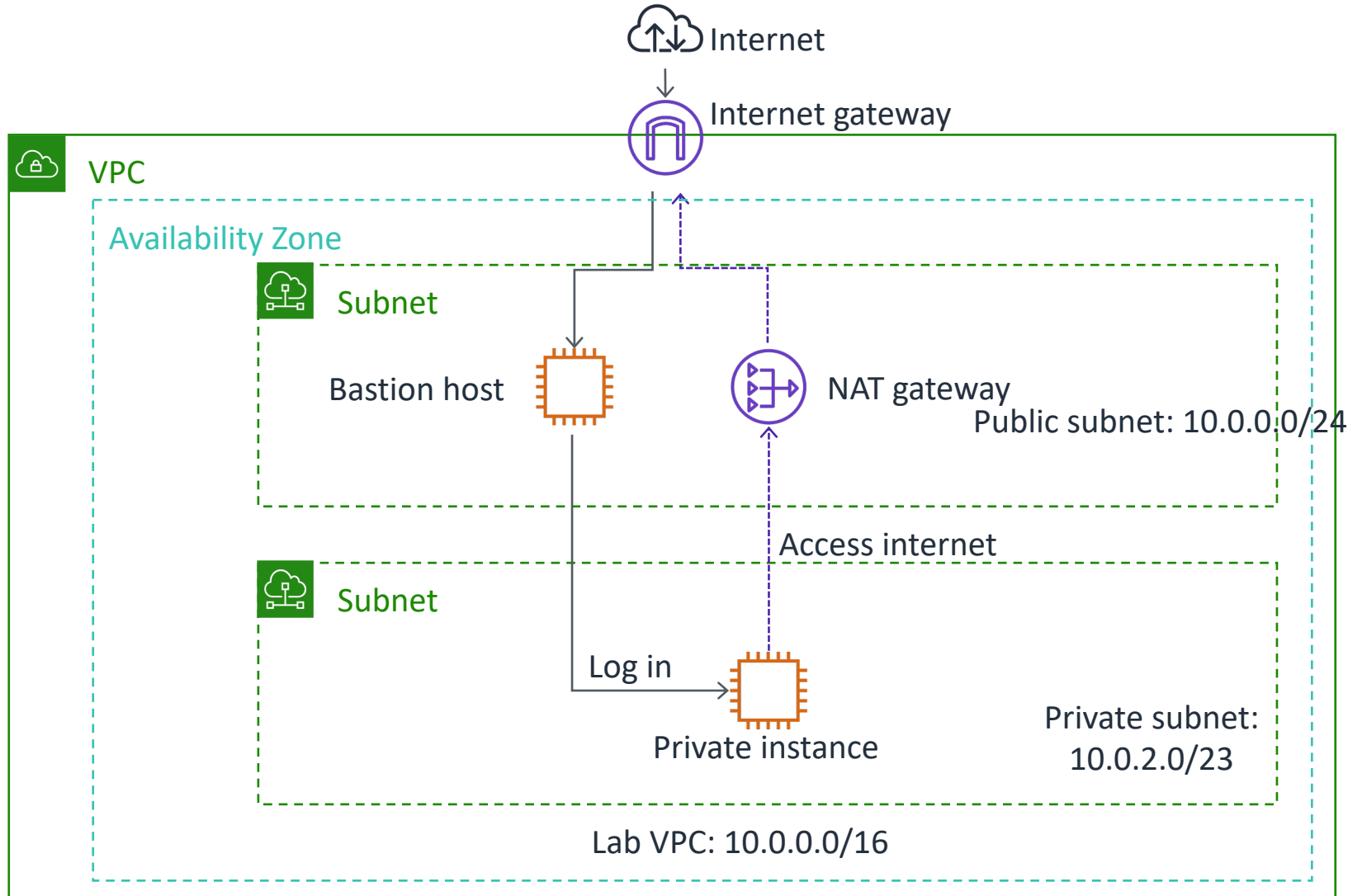
What needs to be updated after establishing a network address translation (NAT) gateway for the unmanned vehicle application so operators can connect and retrieve location data securely? Select three responses.

- A. Implement redundancy in different Availability Zones
- B. Provide access through relevant security groups
- C. Specify a public subnet for creation
- D. Specify an Elastic IP address for association
- E. Update the route table

Lab 4: Configure an Amazon Virtual Private Cloud



Configure an Amazon VPC



Public Route Table

Destination	Target
10.0.0.0/16	Local
0.0.0.0/0	Internet gateway

Private Route Table

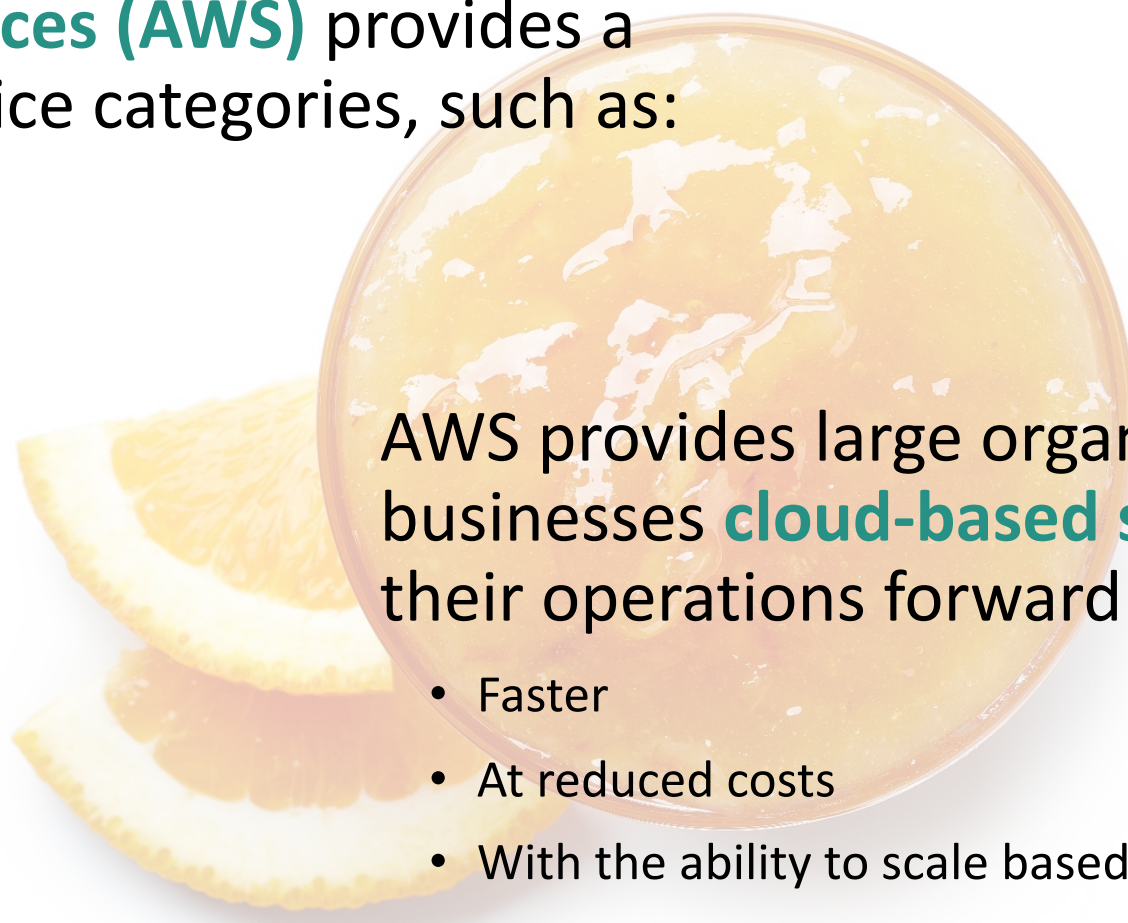
Destination	Target
10.0.0.0/16	Local
0.0.0.0/0	NAT gateway

Systems Operations Project Scenario

Mom & Pop Café & Bakery

Amazon Web Services (AWS) provides a broad array of service categories, such as:

- Compute
- Storage
- Database
- Analytics

A background image featuring a glass of orange juice with a slice of orange floating in it, and two more orange slices on the surface below the glass. The image is semi-transparent, allowing the text to be visible over it.

AWS provides large organizations and small businesses **cloud-based solutions** to move their operations forward:

- Faster
- At reduced costs
- With the ability to scale based on demand

On completion of this project, you will be able to:

- **Experience** the communication challenges faced when attempting to apply technology as the solution to business problems.
- **Translate** customer requirements into a proposed technical solution.
- **Present** the proposed solution to the customer.
- **Use** AWS Best Practices for a frugal, resilient, and secure solution.



Customer Meeting Preparation

Mom & Pop Café & Bakery

Troubleshoot a VPC

Business Need: Troubleshoot the VPC

Sophie is explaining to Emma that recent changes to the Amazon Virtual Private Cloud (Amazon VPC) configuration resulted in:

- Improved network security
- Customers not able to access the website
- The SSH connection to the web server EC2 instance being misconfigured
- The need to download, extract, and analyze the Amazon VPC flow logs



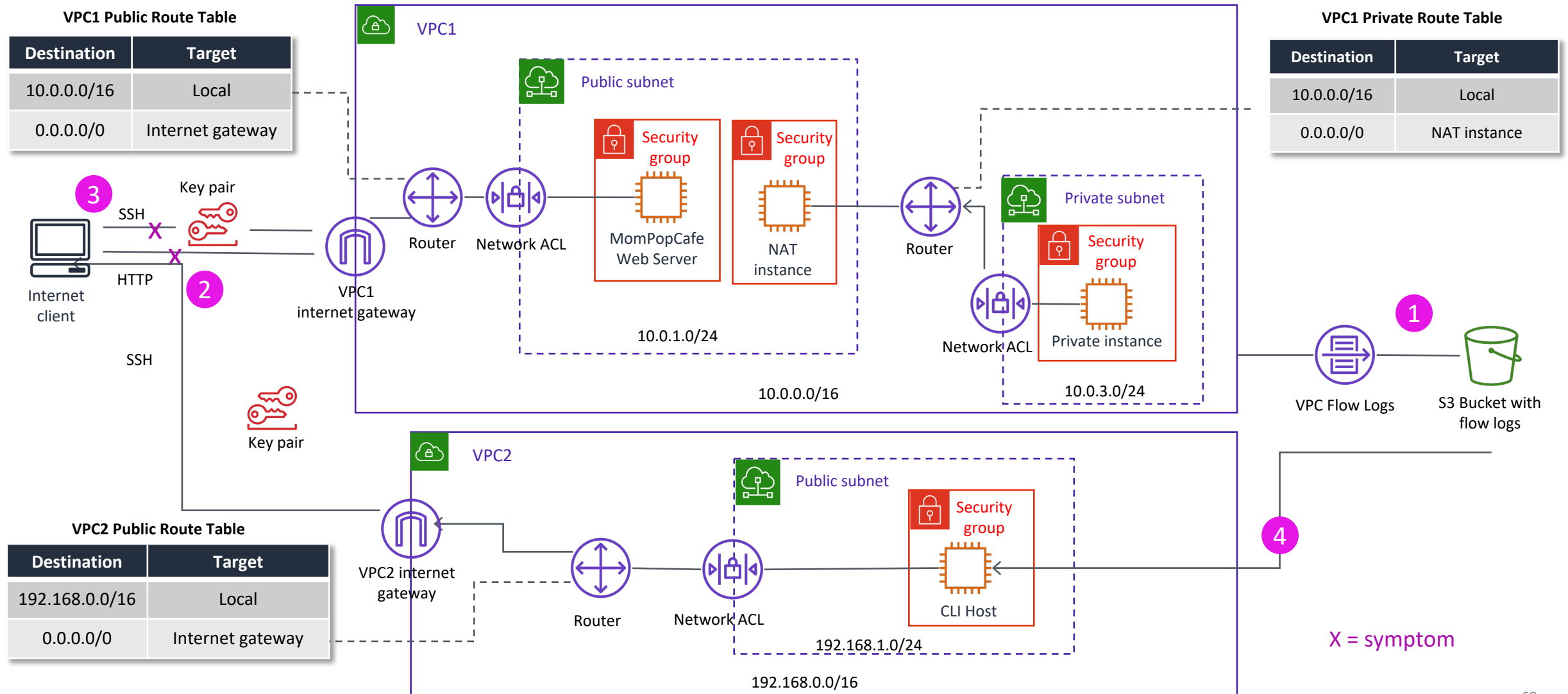
Sophie's Troubleshooting Request

Sophie will need to work through these steps:

1. Configure the AWS CLI on the CLI Host EC2 Instance
2. Enable VPC Flow Logs
3. Analyze and troubleshoot access to resources
4. Download and extract the VPC Flow Logs



Troubleshoot a VPC





Begin Activity #7: Troubleshoot a VPC

Troubleshooting tasks

What are four common troubleshooting tasks that we discussed today?

Answer, but not limited to:

Troubleshooting tasks

What are four common troubleshooting tasks that we discussed today?

Answer, but not limited to:

1. Verify that the instance is up and running and check that it has passed both the System Status and Instance Status checks.
2. Verify that the security groups associated with the instance allow connections for the required protocols and ports.
3. Verify that the network ACLs associated with the subnet allow traffic from the necessary ports and protocols.
4. Verify that the route table associated with the subnet has destination rules that point to the appropriate targets.

Elastic network interface

What are the maximum number of elastic network interfaces that can be added to an EC2 instance?

Answer

Elastic network interface

What are the maximum number of elastic network interfaces that can be added to an EC2 instance?

Answer

The number of network interfaces that can be added to an EC2 instance is dependent on the instance class.

Project time: Troubleshooting knowledge base



Thank You

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